Rival-Conditioned Evidence Attention for Dog-Breed Recognition on Frozen Features: Method, Leakage Audit and Pre-Registered Protocol

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Abstract

Dog-breed recognition leaves many of its errors among a few look-alike breeds, and the local evidence that separates them depends on which breeds are in doubt for the image at hand. We describe rival-conditioned evidence attention (RCEA), a small head on frozen DINOv2 ViT-B/14 features with registers. A global linear head proposes a shortlist of $K = 5$ breeds without gradient. Each candidate forms a query from its class embedding contrasted against the other shortlisted breeds (uniformly, weighted by the stage-1 belief, or pair by pair), attends over PCA-reduced patch tokens, and adds the pooled evidence, scored along the same query, to its own logit. Three controls with exactly the same parameters change only the query (no contrast, contrast with all classes, contrast with random classes); a linear probe, an attentive probe and class attention over all classes are the baselines. Four short facts locate the mechanism among existing heads: uniform rivals are shortlist centring, belief-weighted rivals with tied weights are a log-softmax gradient, the score is a soft maximum of a per-patch margin, and the pairwise variant coincides with the uniform one only at $K = 2$. Tsinghua Dogs is the primary dataset and Stanford Dogs the secondary. A leakage audit finds 19,081 confirmed Stanford copy pairs inside Tsinghua Dogs, covering 3,115 of its 5,200 test images, and finds every Stanford Dogs image in the ILSVRC-2012 training list; results are therefore also reported on provenance subsets fixed by the audit. Learning rates were chosen on validation data only, the selection was frozen with checkpoint hashes, and seven hypotheses with a decision rule over three seeds were fixed before any test evaluation. All training ran on free-tier T4 GPUs: on Colab, and for the secondary studies on Kaggle, which resumed the saved runs after a validation-only check of the rebuilt features. Under that rule the main hypothesis (H1) is refuted: on Tsinghua Dogs the best proposed head, rival-soft, is ahead of the best equal-parameter control, the class query, by +0.41 percentage points, positive in all three seeds but with only one paired interval excluding zero. The rule supports three narrower hypotheses on Tsinghua Dogs: the second stage fixes more test images than it breaks (1.91 % against 0.68 %) and gains most on the most confused breed pairs (H2), random rivals score no higher than centring (H3), and rival-soft beats class attention over all classes by +0.84 points (H5). The pairwise variant shows no difference (H4, exploratory), and robustness is reported as exploratory (H7). None of these effects appears on Stanford Dogs. The rival head's test top-1 peaks at a shortlist of $K = 3$ and falls slowly up to $K = 10$ while the shortlist's recall rises, which meets the criterion of H6, but every step lies within the seed spread and the class query falls from $K = 2$ on. Across four frozen backbones the rule never separates rival from the class query. ConvNeXt-T gives the highest accuracy (87.89 %), although its labelled ImageNet pre-training included 3,115 of the 5,200 Tsinghua test images. On 30 held-out breeds the re-rankers detect unknown breeds about as well as the linear probe.

1 Introduction

Dog-breed recognition is a fine-grained task in which many errors involve a few look-alike breeds. Norfolk and Norwich terriers differ mainly in ear carriage; the Siberian husky and the Alaskan

malamute mainly in size and build; the toy, miniature and standard poodle in size cues that a tight crop removes. Which local evidence decides such a case depends on which breeds are in doubt for the image at hand. A head that pools the patch tokens once, in the same way for every class (mean pooling, or a fixed set of attention queries), cannot make its pooling depend on that set.

The study starts from a premise and measures it instead of assuming it: a strong frozen descriptor puts the correct breed among its first few guesses for most images, and the remaining errors fall inside small, image-specific sets of breeds. If the premise holds, a small second stage that looks at the patches again, conditioned on the image's own shortlist, has something to work on. Its effect can be measured directly: it moves only the K shortlisted logits, the recall of the shortlist says how often the truth is among them, and a changed decision either fixes an error, breaks a correct answer, or swaps one wrong answer for another. These quantities are reported for every re-ranking head (Section 6.3), whatever their values.

We study rival-conditioned evidence attention (RCEA), a head on frozen DINOv2 ViT-B/14 features with registers [1, 2]. A global linear head gives the stage-1 logits z^0 and, without gradient, a shortlist S of $K = 5$ breeds. For each candidate $c \in S$ a query is formed from a learned class embedding contrasted against the other shortlisted breeds: against their mean (*rival*), against a mean weighted by the stage-1 belief (*rival-soft*), or against each rival separately, with a soft minimum over the resulting scores (*pairwise*). The query attends over PCA-reduced patch tokens, the pooled evidence is scored along the same query, and the score is added to the candidate's own logit (Figure 1).

Every ingredient has a precedent, and we cite the closest ones first. The scoring rule (a class query attends over spatial features, and the pooled evidence, projected back on the query, is added to the logit) is that of CSRA [3]. Shortlisting followed by candidate-conditioned attention and re-scoring appears in SAC [4], Look and Think Twice [5] and PCNN [6], always with class-specific queries. The idea that what separates a class depends on which rivals are in doubt is stated by FuDD [7] for zero-shot CLIP, through language descriptions. The uniform and the probability-weighted contrasts are those of the explanation methods CLRP [8] and SGLRP [9]. Our literature search (Section 2) found no work that combines four steps: an image-specific top-K shortlist; one query per candidate contrasted against the other $K-1$; cross-attention with that query over patch tokens; and re-scoring along the same query, added to that candidate's logit. The mechanism is therefore new as a specific construction and incremental as an idea. Fact 1 makes the honest form of the claim precise: the uniform-rival query is the class embedding centred on the shortlist, so the experiments ask whether it matters where the centring set comes from.

All heads are trained on cached features of frozen backbones. The choice follows from the compute that was available: one free-tier Colab T4 GPU (15 GB) with 2 virtual CPUs and 12 GB of RAM, later two free-tier Kaggle T4 GPUs, and nothing paid. It also keeps the comparisons between heads clean, since all heads of an experiment read identical features. End-to-end fine-tuning of the backbone is not part of the study, and whether any effect of the head survives it stays open.

Two public datasets are used: Tsinghua Dogs [10] (130 breeds, primary) and Stanford Dogs [11] (120 breeds, secondary). Both carry leakage that changes how their numbers can be read. Tsinghua Dogs integrates about 18,000 Stanford Dogs images [10], and Stanford Dogs was built from ImageNet [11, 12]. Our audit (Section 3) finds a confirmed Stanford copy for 3,115 of the 5,200 Tsinghua test images and finds every Stanford Dogs image in the ILSVRC-2012 training list. Absolute accuracies are therefore optimistic for every ImageNet-trained backbone, and we also report on the Tsinghua test images that match no Stanford image at all.

The contributions are stated as what is tested, not as outcomes.

1. RCEA and four short facts that locate it among existing heads: uniform rivals are shortlist centring; belief-weighted rivals with tied weights reduce to a log-softmax gradient, which is

why the embeddings are untied; the score is a soft maximum of a per-patch margin; and the pairwise mode equals the rival mode at $K = 2$ but is not a single query for $K \geq 3$ (Section 4).

2. A single-variable control set. Three controls with exactly the parameters of the proposed heads change only the query (no centring, centring on all classes, centring on random classes), next to three baselines (linear probe, attentive probe, class attention over all classes; Table 8).

3. A leakage and provenance audit of Tsinghua Dogs against Stanford Dogs and of Stanford Dogs against ILSVRC-2012, with provenance subsets of the Tsinghua test split fixed before any test evaluation (Section 3).

4. A pre-registered protocol: learning rates selected on validation data only and frozen with checkpoint hashes, seven falsifiable hypotheses, and a decision rule over three seeds with paired per-image tests (Section 5).

The test results are in Section 6; each table and figure there is generated from the result digest by scripts in the repository. The Colab GPU quota ended before the secondary studies were finished (the shortlist-size and training-choice ablations, the token widths, the open set, two further backbones and the cost measurements). They were finished on Kaggle under the same protocol, from the runs saved on Google Drive, after a check fixed in advance had shown that the rebuilt feature caches change at most 3 of the 6,515 validation predictions of any trained head (Section 5.3).

The results do not support the central claim. On Tsinghua Dogs, rival-soft, the proposed mode that validation selected, reaches 87.12 % test top-1 against 86.71 % for the class query, the best control; the difference is positive in all three seeds (+0.37, +0.50 and +0.37 points), but its paired interval excludes zero in only one of them, so the rule makes no claim and H1 is refuted. The verdict is the same on the 1,362 test images that match no Stanford Dogs image (+0.69 points, no claim). The rule supports narrower statements. The second stage of rival-soft fixes 1.91 % of the test images and breaks 0.68 %, and its gain over the linear probe is larger on the breeds of the 20 most confused pairs than overall (+1.34 against +0.86 points; H2). Random rivals score no higher than centring (H3). Rival-soft beats class attention over all 130 classes (+0.84 points; H5) and the linear probe (+0.86 points), and both leads hold on the independent images (+1.39 and +1.32 points). The pairwise variant is indistinguishable from the uniform one (H4), and the lead over the linear probe grows under blur, low resolution and occlusion (H7, exploratory). On Stanford Dogs, whose images are all ILSVRC-2012 training images, no re-ranking head differs from the linear probe by the rule, and every mean difference to it is negative. The secondary studies do not change this picture. Shortlists of $K = 2$, 3 and 10 give the rival head 86.85, 86.91 and 86.52 % (86.74 % at $K = 5$), and no training choice moves it by more than 0.15 points. On CLIP ViT-B/16 and ConvNeXt-T features both the class query and rival beat the linear probe by the rule, but, as on DINOv2 and ResNet-50, the rule does not separate the two. With 30 breeds held out, every head detects them about equally well (AUROC of the maximum softmax probability 80.6–80.9 %).

2 Related work

Fine-grained recognition and dog breeds. Fine-grained visual classification (FGVC) has been approached with bilinear pooling [13], part and region discovery [14–16], multi-granularity and destruction–construction training [17, 18], channel and attention regularisers [19–22], and vision transformers with token selection, fusion or cross-attention [23–33]. Dog breeds were studied early with part localisation [34] and on the Stanford Dogs benchmark [11]. Four recent works frame this study. Tsinghua Dogs [10] has 70,428 images of 130 breeds, at least 200 per breed, with semi-automatic head and body boxes (RetinaNet proposals corrected by annotators). Its official split has 65,228 training and 5,200 validation images (40 per breed) and no separate

test set, and about 18,000 of its images were taken from Stanford Dogs (their Sec. 3.1). Its benchmark (their Table 3) gives 77.66 % for Inception V3 at 224×224 and 86.404 % for WS-DAN at 512×448. The Head-Body Guided framework [35] detects head and body with YOLOv8n and classifies the two crops with two ViT-B/16 streams at 224 px. It reports 90.04 % on Tsinghua Dogs (their Sec. 4.4.2), but states neither the evaluation split nor whether the evaluated crops come from the ground-truth boxes or from the detector. CSQA-Net [36] assesses feature quality with stop-gradient probes and adds a multi-part cross-attention branch that is removed at test time. It reports 90.5 and 92.6 % on CUB-200-2011 with ResNet-50 and Swin-B, evaluates on no dog dataset, and its cross-attention queries are class-agnostic. Attentive batch training (RBI) [37] lets every test image attend to the other images of its test batch, so its predictions are transductive. Its Stanford Dogs number is 95.79 % in the arXiv version (Table 1) and 97.79 % in the journal abstract; we could not resolve the discrepancy. Table 1 collects reported numbers. None of them is comparable with this study, whose backbone is frozen.

Class-specific queries and attention heads. CSRA [3] computes, for every class, a spatial softmax attention from that class's classifier weight, pools a class-specific feature, and adds its inner product with the same weight to the logit; with an infinite attention temperature it becomes average plus max pooling. Query2Label [38] and ML-Decoder [39] decode class or group queries against the feature map; MCTformer [40] uses one class token per class; AttentionXML [41] attends per label after shortlisting labels with a tree; INTR [42] gives each class a query in a DETR-style decoder for interpretability (72.5 % on Stanford Dogs against 77.1 % for ResNet-50, their Table 2). In these heads the queries are the same for every image and cover all classes, or a label-tree shortlist, and none is contrasted against competing candidates. Attentive probes on frozen encoders pool the tokens with learned class-agnostic queries: CAE [43], V-JEPA [44], AIM [45] and Efficient Probing (EP) [46]. On ImageNet-1k with DINOv2 ViT-B/14, EP reports 81.8 % for k-NN, 83.2 % for a linear probe and 84.0 % for EP (their Table 2). Our baseline B1 is an attentive probe of this kind, and B2 is CSRA's form over all classes with learned projections.

Shortlists and re-ranking. Re-scoring a classifier's top candidates is an old idea. Look and Think Twice [5] derives one feedback attention box per top-5 label and re-classifies the crops, which lowers the ImageNet-2014 validation top-1 error by 1.79 points (their Table 3). The specialists of Hinton et al. [47], the Network of Experts [48] and HD-CNN [49] route images to models trained on confusable class subsets fixed in advance. SAC [4] is the closest pipeline: the label embeddings of the top-k ($k = 10$) candidates attend over the feature map through one bilinear attention, a second classifier re-predicts from the jointly pooled vector, and the two predictions are averaged. It reports an average gain of 1.2 points on Stanford Dogs over seven backbones and, in a k ablation on CUB, 87.2, 88.4, 88.3, 87.7 and 85.9 % for $k = 2, 5, 10, 20, 50$. PCNN [6] multiplies each top-K confidence by a learned comparison with that class's nearest training image (Stanford Dogs with ResNet-50: 85.82 to 86.31 %), and CCFR [50] re-ranks with retrieval statistics. Shortlists also feed language models: CascadeVLM [51] and RAR [52] pass CLIP's top-k classes to a large (multimodal) language model. Adaptive-computation methods such as GFNet [53] and AdaFocus [54–56] take a second, focused look where the first is uncertain. In these works each candidate is examined with its own, class-specific signal; none contrasts a candidate against the other members of the image's shortlist.

Contrastive explanations and rival-dependent descriptions. The contrast between a class and its competitors is established in explanation methods, where it is used post hoc and is never trained or fed back into the prediction. The contrastive map of Excitation Backprop [57, 58] subtracts the map of a dual unit with negated weights. CLRP [8] subtracts the relevance of a virtual class formed from all other classes (uniformly) or from the negated weights. SGLRP [9] starts from the softmax gradient, which weights every other class by its probability. Contrastive

explanations in the sense of "why P, rather than Q" [59] use one chosen contrast class; SCOUT [60] and counterfactual visual explanations [61] highlight evidence for a class against a counter class or a distractor image. In RCEA's terms, CLRP's uniform virtual class is the global-centring control and SGLRP's weighting is the belief-weighted query (Fact 2). On the language side, FuDD [7] forms, for each image, the set of its k most ambiguous classes and has a language model describe the visual differences of every pair; on Stanford Dogs it reports 60.03 % (CLIP ViT-B/32) and 75.99 % (ViT-L/14@336) zero-shot with $k = 10$. Comparative descriptors [62] fix the rivals per class in advance, and classification by description [63] uses no rivals. POOF [64] learns part-based features for every pair of classes. RCEA's pairwise mode is the learned, visual counterpart of FuDD's pair descriptions and of POOF's pair features, formed per image.

Confusion-aware learning. Several objectives use the confusion structure during training only. Pairwise Confusion [65] and maximum-entropy training [66] regularise the output distribution; online label smoothing [67] builds soft labels from the model's own predictions; Seesaw loss [68] rebalances the penalties between classes; the gradient-boosting cross-entropy of Sun et al. [69] normalises the softmax over the true class and the top-k negatives; Sharpen Focus [70] penalises attention shared with the most confusing class; FocusNet [71] trains a second network on each image's confusing classes; API-Net [72] exchanges gated features between each image and its most similar images in the batch, of the same and of other classes, during training only. Few-shot methods condition class representations on the classes of an episode (FEAT [73], CTM [74]) or cross-attend between a query image and each support class (CAN [75], CrossTransformers [76]). In none of them does the test-time computation depend on the image's own shortlist.

Frozen backbones and probing. Self-supervised and language-supervised backbones are routinely evaluated on frozen features: DINOv2 [1] with a linear head on the class token concatenated with the mean patch token, DINOv3 [77], I-JEPA [78] with a linear probe, and CLIP [79]. Registers [2] remove artefact tokens, which appear on about 2 % of the patches with about ten times the norm of the others; this matters for any head that attends over patch tokens. On Stanford Dogs, VPT [80] reports, for a frozen supervised ImageNet-21k ViT-B/16 at 224 px, 86.2 % for a linear probe, 91.2 % for bias tuning, 90.7 % for VPT-shallow and 89.4 % for full fine-tuning (their Table 14), so in that setting several variants that keep the backbone frozen score above full fine-tuning. Supervised ImageNet backbones (ResNet [81], ConvNeXt [82], EfficientNet [83], ViT [84]) complete the picture; two of them are secondary backbones here.

Pre-training overlap. Stanford Dogs is a subset of the ILSVRC training images, so Krause et al. [12] pre-train on ILSVRC with the dog data removed (their Sec. 5.1); they report 80.6 % with the Stanford training images alone and 85.9 % with added web images and multi-crop testing. Murabito et al. [85] and MetaFGNet [86] make the same point; MetaFGNet's ResNet-34, pre-trained without the 120 dog classes, reaches 69.3 % (73.2 % with their method). DINOv2's LVD-142M contains ImageNet-22k as is (14,197,086 images) plus images retrieved with ImageNet-22k and the ImageNet-1k training split as seeds (their Table 15), and Stanford Dogs is not among its de-duplication targets. CLIP's overlap analysis over 35 datasets, which do not include Stanford Dogs, finds a median overlap of 2.2 % and a largest accuracy gain of 0.6 % [79]. BiT [87] lists 58 Oxford-IIIT Pet [88] test images with a near-duplicate in the ILSVRC-2012 training set, and ImageNetV2 [89] offers new test images for the ImageNet classes, with near-duplicates of ImageNet removed. Section 3.5 applies these sources to the backbones used here.

Calibration, selective prediction and open set. We report temperature-scaled NLL and the expected calibration error [90, 91], risk–coverage summaries of selective classification [92], and open-set detection with the maximum softmax probability [93], the energy score [94] and the maximum logit, which Vaze et al. [95] recommend as a baseline when the closed-set classifier

is strong. Their Semantic Shift Benchmark has no dog split, so we hold out breeds of Tsinghua Dogs instead (Section 5.9).

Explanations and naming. Prototype models (ProtoPNet [96], PIP-Net [97]) and correspondence-based classifiers (CHM-Corr [98]) explain fine-grained decisions through patches; we only measure where the attention of each head falls relative to the annotated boxes (Section 5.14). Three further works share a single ingredient with RCEA: top-down attention by analysis by synthesis (AbSViT [99]), training-free region scoring with CLIP margins [100], and visual-token pruning for re-rankers that discounts patterns shared across candidates [101], the only other construction we found that scores something relative to the other candidates, in retrieval rather than classification. We avoid the name "differential attention", which denotes a difference of two softmax maps in the Differential Transformer [102], and "contrastive attention", a term of Excitation Backprop [57].

Table 1: Reported top-1 accuracy (%) of related methods, copied from `results/literature_table.csv` (Stanford Dogs) and from the dossier `research_log/lit_core.md` (Tsinghua Dogs), with the table or section of the paper each number comes from. Backbone, pre-training, resolution, fine-tuning and inference mode differ between rows, so the rows are comparable neither with each other nor with this study. Section 3.5 explains why every ImageNet-pretrained row on Stanford Dogs is optimistic. IN-1k / IN-21k: ImageNet-1k / -21k pre-training.

Method	Venue	Backbone	Pre-training	Input	Reported top-1 (where in the paper)
Stanford Dogs, fine-tuned backbones					
WS-DAN [15]	arXiv 2019	Inception-v3	IN-1k	not stated	92.2 (Table 7; Inception-v3 baseline 88.9)
API-Net [72]	AAAI 2020	R50 / R101 / DN-161	IN-1k	512, crop 448	88.3 (ResNet-50), 90.3 (ResNet-101), 89.4 (DenseNet-161) (Table 9)
PC [65]	ECCV 2018	DN-161 / B-CNN / R50	IN-1k	not stated	83.75 (PC-DenseNet-161; baseline 81.18); 83.04 (PC-Bilinear CNN; baseline 82.13); 73.35 (PC-ResNet-50; baseline 69.92); Table 2(F), mean of 5 trials
MaxEnt [66]	NeurIPS 2018	DN-161 / B-CNN / R50	IN-1k	not stated	83.63 (MaxEnt-DenseNet-161; baseline 81.18); 83.18 (MaxEnt-Bilinear CNN; 82.13); 73.56 (MaxEnt-ResNet-50; 69.92); Table 1(E), mean of 6 trials
LGTF [103]	ICCV 2023	DN-161 / R101	IN-1k	448	92.1 (DenseNet-161), 91.7 (ResNet-101) (Table 3); their plain ResNet-101 baseline 80.6, DenseNet-161 80.9
Relational Proxies [104]	NeurIPS 2022	R50	IN-1k	not checked	92.75 (Table 4; their relation-agnostic ResNet-50 encoder 91.30)
CAP [105]	AAAI 2021	Xception	IN-1k	256, crop 224	96.1 (Table 1)
SR-GNN [106]	TIP 2022	Xception	IN-1k	256, crop 224	97.3 (Table I)
TransFG [23]	AAAI 2022	ViT-B/16	IN-21k	448	92.3 (Table 2; plain ViT-B/16 91.7)
FFVT [24]	BMVC 2021	ViT-B/16	IN-21k	600, crop 448	91.5 (Table 3; their re-run of ViT-B/16 90.2 and of TransFG 90.6, against TransFG's own 92.3)
RAMS-Trans [25]	ACM MM 2021	ViT-B/16	IN-21k	short side 448, crop 224	92.4 (Table 3; ViT-B/16 92.2; their TransFG&PSM re-run 90.0)
MP-FGVC [31]	AAAI 2024	ViT-B/16 + CLIP text	IN-21k	448	91.0 (Table 1); their re-runs: TransFG* 90.5, IELT* 90.6, SIM-Trans* 89.2
ViT-NeT [33]	ICML 2022	DeiT-B / Swin-B	IN-1k (DeiT) / IN-22k (Swin)	448	93.6 with DeiT-B, 90.3 with Swin-B (Table 2); bases: DeiT-B 91.5, Swin-B 88.0
RBI [37]	ICT Express 2026	ConvNeXt-L and others	IN-1k	224	95.79 (arXiv v1, Table 1; ConvNeXt-Large baseline 93.71) or 97.79 (journal abstract); transductive batch inference
Stanford Dogs, frozen or zero-shot backbones, and pre-training without dog data					
VPT [80]	ECCV 2022	frozen ViT-B/16	supervised IN-21k	224	linear probe 86.2, MLP-2 85.4, partial-1 85.5, bias 91.2, adapter-256 89.9, VPT-shallow 90.7, VPT-deep 90.2, full fine-tune 89.4 (Table 14)
FuDD [7]	ICLR 2024	frozen CLIP	WIT-400M	224 / 336	zero-shot 60.03 (B/32) / 75.99 (L/14@336) with $k = 10$; 60.28 / 76.34 with $k = \|C\|$; single template 58.01 / 73.61; Menon & Vondrick descriptors 57.86 / 74.02 (Table 1)
Krause et al. [12]	ECCV 2016	BN-Inception	ILSVRC with dog data removed	–	80.6 (Stanford-GT only); 80.8 (L-Dog web data, multi-crop); 84.0 (L-Dog + Stanford-GT); 85.9 (L-Dog + Stanford-GT, multi-crop) (Table 1)
Tsinghua Dogs (official split, 5,200 validation images used as test)					
Zou et al. [10]	CVM 2020	Inception V3	not stated	224	77.66 (Table 3)
WS-DAN, ibid.		Inception	not stated	512×448	86.404 (Table 3)
PMG, ibid.		ResNet-50	not stated	448 crop of 550	83.52 (Table 3)
TBMSL-Net, ibid.		ResNet-50	not stated	448 + 224	83.7 (Table 3)
Head-Body Guided [35]	CMC 2025	2×ViT-B/16 + YOLOv8n	ImageNet, variant not stated	224	90.04 (Sec. 4.4.2, Table 6); one stream on the original image 82.54 (Table 3); split and crop source not stated

3 Datasets and leakage audit

3.1 Sources and terms

Tsinghua Dogs [10] was downloaded from the official URLs: the low-resolution release (2.66 GB), the official split files and the annotations. Its authors make it available for academic purposes. The official host of Stanford Dogs did not answer, and image-net.org was unreachable as well. The download script therefore fetched Internet Archive snapshots of the official URLs and accepted a file only if its SHA-256 equalled the checksum that TensorFlow Datasets records for the official release. All three archives matched (Appendix B), so the bytes are those of the official release and only the transport differs. Stanford Dogs has no licence of its own and inherits the ImageNet terms, which allow non-commercial research and educational use only [11, 107].

3.2 Splits and within-dataset audit

Every image was decoded and hashed: SHA-1 for exact duplicates, and a 64-bit DCT perceptual hash (pHash) and a difference hash for near duplicates. Training images that duplicate a test image exactly or nearly (pHash Hamming distance at most 4) are excluded from training. A stratified 10 % of the official training images (seed 0) forms the validation split used for all model selection. The test splits are the official ones: for Tsinghua Dogs, the 5,200-image official validation split (40 images per breed), which is that benchmark's de facto test set; for Stanford Dogs, the 8,580 official test images.

The research log records the following counts (Table 2 is generated from the same audit files). Tsinghua Dogs has 58,627 training, 6,515 validation and 5,200 test images; 86 training images were excluded as duplicates of test images; no file was undecodable; there are 114 groups of exact duplicates, 700 near-duplicate pairs (87 of them between training and test) and 113 near pairs whose labels differ. Stanford Dogs has 10,728 training, 1,199 validation and 8,580 test images, 73 excluded training images, no undecodable file, 89 exact groups, 156 near pairs (74 between training and test) and 48 near pairs with different labels. The label conflicts were recorded with examples and not relabelled, since nothing could arbitrate between the two labels. The Tsinghua data are long-tailed (at least 200 and at most 7,449 images per breed in the whole dataset [10]), while its test split is balanced.

Table 2: Within-dataset audit. For each dataset: the images listed; the splits used (validation is a stratified 10 % of the official training split); the images left out of every split, and among them the training images that duplicate a test image; undecodable files; groups of exact duplicates (SHA-1); near-duplicate pairs (pHash Hamming distance at most 4; pairs between training and test in brackets); and near pairs whose labels differ.

		Split used				Removed		Duplicates		
Dataset	Images	Train	Val	Test	Excl.	Train dup. of test	Undecodable	Exact groups	Near pairs (train–test)	Label conflicts
Tsinghua Dogs	70,428	58,627	6,515	5,200	86	86	0	114	700 (87)	113
Stanford Dogs	20,580	10,728	1,199	8,580	73	73	0	89	156 (74)	48

3.3 Stanford Dogs inside Tsinghua Dogs

Tsinghua Dogs stores the Stanford photographs at another resolution and encoding, so byte hashes cannot find them. The audit (`scripts/cross_overlap.py`) compares every Tsinghua image with every Stanford image on two independent signals: the Hamming distance of their pHashes, and the cosine similarity of their frozen DINOv2 global descriptors g (same backbone and view for both datasets). A pair is a *candidate* if its pHash distance is at most 12 or its cosine at least 0.80. It is *confirmed* if both signals agree (pHash distance at most 8 and cosine at least 0.85) or if the cosine alone is at least 0.97. The images excluded by the splits have no cached descriptor; they keep the pHash signal, so they can be candidates but never confirmed.

The inputs are 70,428 Tsinghua and 20,580 Stanford images, of which 70,342 and 20,507 have descriptors. The audit finds 155,728 candidate pairs and 19,081 confirmed pairs. The confirmed pairs cover 18,675 Tsinghua images and 18,578 Stanford images (90.3 % of Stanford Dogs), in line with the about 18,000 Stanford images that Zou et al. [10] report integrating. 98.8 % of the confirmed pairs carry the same breed name. 120 of the 130 Tsinghua breeds have confirmed copies, and within each of them the majority Stanford breed accounts for at least 96.2 % of the pairs (median 100 %). By split (Tsinghua / Stanford), the confirmed pairs are: training / training 8,280; training / test 6,003; test / training 1,869; test / test 1,305; validation / training 946; validation / test 678. Of the 8,580 Stanford test images, 6,482 have a confirmed copy in the Tsinghua training or validation data, and 973 have no candidate there at all.

Threshold sensitivity. Applying the two-signal clause alone with pHash thresholds 4, 6, 8, 10 and cosine thresholds 0.80, 0.85, 0.90, 0.95, the number of matched Tsinghua images stays between 18,613 and 18,664; the full rule, which adds the cosine-only clause, gives 18,675.

Visual check. Random pairs from each similarity band were rendered as thumbnails and inspected. All 4 inspected pairs with cosine at least 0.97 (a band of 18,991 pairs) and both inspected pairs with pHash distance at most 8 and cosine between 0.90 and 0.97 (88 pairs) show the same photograph, once with an added watermark and once converted to greyscale. Both inspected pairs with pHash distance at most 8 but cosine below 0.85 (151 pairs, only 13.2 % with the same breed) show different dogs, i.e. pHash collisions; this is why the rule needs both signals. Of 6 inspected pairs with pHash distance above 8 and cosine between 0.90 and 0.97 (4,157 pairs, 1,644 Tsinghua images), at least 2 show the same photograph under a different crop, which pHash does not tolerate. The confirmed set therefore has high precision, but it is a lower bound on the Stanford-origin images.

Table 3: Cross-dataset audit of Tsinghua Dogs against Stanford Dogs: candidate and confirmed pairs, the images they cover, the share of confirmed pairs with the same breed name, the Stanford test images confirmed in Tsinghua training or validation data, the clean Stanford test list used for transfer (no candidate in Tsinghua training or validation data), and the confirmed pairs by split (Tsinghua split / Stanford split). A pair is confirmed when its pHash Hamming distance is at most 8 and the cosine of the two DINOv2 global descriptors at least 0.85, or when the cosine alone is at least 0.97 (Section 3.3).

Cross-dataset audit (Tsinghua vs Stanford)	Count
Candidate pairs	155,728
Confirmed pairs	19,081
Tsinghua images with a confirmed Stanford copy	18,675
Stanford images with a confirmed Tsinghua copy	18,578
Confirmed pairs with the same breed name	98.8%
Stanford test images confirmed in Tsinghua train/val	6,482
Clean Stanford test images (no candidate in Tsinghua train/val)	973
Confirmed pairs by split:	
Tsinghua test / Stanford test	1,305
Tsinghua test / Stanford train	1,869
Tsinghua train / Stanford test	6,003
Tsinghua train / Stanford train	8,280
Tsinghua val / Stanford test	678
Tsinghua val / Stanford train	946

3.4 Consequences and provenance subsets

Three consequences were drawn before any test evaluation. (i) The Tsinghua test split is far richer in Stanford images than its training data (Table 4): 3,115 of the 5,200 test images (59.9 %) have a confirmed Stanford copy and 3,838 (73.8 %) have at least one candidate, against 1,584 of 6,515

validation images (24.3 %) and 13,976 of 58,627 training images (23.8 %) with a confirmed copy. (ii) The test split is divided once, by the audit alone, into provenance subsets: *Stanford-origin* (a confirmed copy; 3,115 images), *independent* (no candidate pair at all; 1,362 images), and the remaining 723 candidate-only images, which are counted but not analysed. Because the confirmed set is a lower bound, the independent subset excludes every candidate, not only the confirmed ones. It is not a random sample: it holds fewer typical Stanford-like photographs and includes the 10 Tsinghua breeds with no confirmed Stanford copy. Comparisons between heads on it are informative; its absolute accuracy, set against the full test split, is not. (iii) Training on one dataset and testing on the other needs de-duplication. The transfer study (Section 5.10) uses only the 973 Stanford test images with no candidate in the Tsinghua training or validation data.

Table 4: Stanford Dogs content of the Tsinghua splits, and the provenance subsets of the Tsinghua test split, fixed by the audit before any test evaluation (counts from the research log). Every image with a confirmed copy is, through that copy, an ILSVRC-2012 training image (Section 3.5).

	Images	Confirmed Stanford copy	At least one candidate
Tsinghua training split	58,627	13,976 (23.8 %)	–
Tsinghua validation split	6,515	1,584 (24.3 %)	–
Tsinghua test split	5,200	3,115 (59.9 %)	3,838 (73.8 %)
test subset “Stanford-origin”	3,115	all	all
test images with a candidate only (not analysed)	723	none	all
test subset “independent”	1,362	none	none

3.5 Pre-training overlap

Stanford Dogs keeps the ImageNet file names, so its overlap with ILSVRC-2012 can be counted by file id (`scripts/ilsvrc_overlap.py`, against the 1,281,167-line training list `train.txt` of the BVLC Caffe ILSVRC-2012 auxiliary archive). All 120 Stanford Dogs classes are ILSVRC-1k classes, and every Stanford Dogs image is an ILSVRC-2012 training image: 12,000 of the 12,000 official training images and 8,580 of the 8,580 official test images. Through their confirmed copies, all 3,115 Stanford-origin Tsinghua test images are ILSVRC-2012 training images too, as are 13,976 Tsinghua training and 1,584 validation images.

For the backbones this means the following. ResNet-50 (torchvision `IMAGENET1K_V2`) and ConvNeXt-T (`IMAGENET1K_V1`) were trained with labels on every Stanford Dogs test image. DINOv2's LVD-142M includes ImageNet-22k as is (14,197,086 images; Table 15 of Oquab et al. [1]), which contains the ImageNet images Stanford Dogs was drawn from; so the Stanford Dogs images were in DINOv2's pre-training data, without labels. This is an inference from the published data composition; we did not count it. CLIP's training data are not public. Absolute accuracies on Stanford Dogs and on the Stanford-origin part of Tsinghua Dogs are therefore optimistic for every backbone used here. All heads of one experiment read the same frozen features, so comparisons between heads are less exposed, but a head could still draw more on memorised images than another; the independent subset is the cleaner check.

4 Method

4.1 Notation

An image is represented by the patch tokens and a global descriptor of a frozen backbone (Table 5). The $N = 256$ patch tokens (a 16×16 grid at 224 px with patch size 14; the four register tokens are dropped) are projected on the first D principal components of the training tokens, giving $X = [x_1, \ldots, x_N]$ with $x_i \in \mathbb{R}^D$. The global descriptor $g = [\mathrm{CLS}; \bar{x}] \in \mathbb{R}^{D_g}$ is the class token concatenated with the mean patch token, as in DINOv2's linear evaluation. LN is

layer normalisation with learned affine parameters, $\mathrm{sg}(\cdot)$ stops the gradient, and C is the number of breeds.

Table 5: Notation and the values of the main study.

Symbol	Meaning	Main study
$X = [x_1, \ldots, x_N]$, $x_i \in \mathbb{R}^D$	frozen patch tokens (registers dropped) on the first D principal components of the training tokens	$N = 256$; $D = 128$ of a 256-component PCA fitted on 4,000 training images
$g \in \mathbb{R}^{D_g}$	global descriptor $[\mathrm{CLS}; \bar{x}]$	$D_g = 1536$
C	number of breeds	130 (Tsinghua), 120 (Stanford)
$z^0 = W\,\mathrm{LN}(g) + b^0$	stage-1 logits	
$S = \mathrm{top}_K(\mathrm{sg}(z^0))$	the image's shortlist	$K = 5$
$e_j \in \mathbb{R}^d$	learned class embedding, untied from W	$d = 256$
H, $d_h = d/H$	attention heads and head width	$H = 4$, $d_h = 64$
$W_k, W_v \in \mathbb{R}^{d \times D}$	key and value projections of $\mathrm{LN}(x_i)$, shared by all queries, no bias	
$b \in \mathbb{R}^d$	one class-agnostic attention bias	
γ	learned scale of the second-stage score	initialised to 1
τ, T, t	attention temperature; rival-soft temperature; pairwise soft-min temperature	1, 1, 1
α	weight of the stage-1 loss	1

4.2 Stage 1 and the shortlist

$$z^0 = W\,\mathrm{LN}(g) + b^0, \qquad S = \mathrm{top}_K\big(\mathrm{sg}(z^0)\big). \tag{1}$$

The shortlist is taken without gradient, so the second stage cannot move candidates into or out of its own shortlist. During training only, if the true class y is not in S, it replaces the weakest candidate (`gt_insert`); the second stage thus always learns to separate the truth from its hardest rivals. At test time S is the top-K of z^0 alone: inference is automatic, one image at a time, without batch statistics and without annotations.

4.3 Queries

Each class j has a learned embedding $e_j \in \mathbb{R}^d$, untied from the stage-1 weights W (Fact 2 gives the reason). For a candidate $c \in S$ the three proposed queries are

$$\text{rival:} \quad q_c = e_c - \frac{1}{K-1} \sum_{r \in S \setminus \{c\}} e_r, \tag{2}$$

$$\text{rival-soft:} \quad q_c = e_c - \sum_{r \in S \setminus \{c\}} w_{c,r}\, e_r, \qquad w_{c,r} = \frac{\exp\big(\mathrm{sg}(z^0_r)/T\big)}{\sum_{r' \in S \setminus \{c\}} \exp\big(\mathrm{sg}(z^0_{r'})/T\big)}, \tag{3}$$

$$\text{pairwise:} \quad q_{c,r} = e_c - e_r \quad \text{for every } r \in S \setminus \{c\}. \tag{4}$$

The rival-soft weights are a softmax over the rivals of the detached stage-1 logits. The pairwise mode forms $K - 1$ queries per candidate, each with its own attention map (Section 4.4). The three controls have exactly the same parameters and change only the query:

$$\text{class: } q_c = e_c, \qquad \text{centre: } q_c = e_c - \frac{1}{C}\sum_{j=1}^{C} e_j, \qquad \text{random: } q_c = e_c - \frac{1}{K-1}\sum_{m=1}^{K-1} e_{\rho_m}, \tag{5}$$

where $\rho_1, \ldots, \rho_{K-1}$ are drawn uniformly, with replacement, from the $C - 1$ classes other than c. The class query is CSRA's form restricted to the shortlist [3]; the centred query is the contrast of

CLRP's uniform virtual class [8]; the random query keeps a contrast but not the image's confusion set. In training, the random classes are drawn anew at every forward pass. At evaluation they come from a generator seeded with the run's seed, so an evaluation repeats exactly, and the spread over five further draws is reported (Section 5.6).

4.4 Evidence attention and score

Keys and values are computed once per image and shared by all queries: $k_i = W_k\,\mathrm{LN}(x_i)$ and $v_i = W_v\,\mathrm{LN}(x_i)$, with $W_k, W_v \in \mathbb{R}^{d\times D}$ and no bias. Queries, keys, values and the bias are split into H heads of width $d_h = d/H$, marked by a superscript (h). With one class-agnostic attention bias $b \in \mathbb{R}^d$ ("look at the dog") and the temperature τ,

$$a_{c,i}^{(h)} = \mathrm{softmax}_i\left(\frac{(q_c^{(h)} + b^{(h)})^\top k_i^{(h)}}{\sqrt{d_h}\,\tau}\right), \tag{6}$$

$$r_c^{(h)} = \sum_{i=1}^{N} a_{c,i}^{(h)}\, v_i^{(h)}, \tag{7}$$

$$s_c = \frac{1}{\sqrt{d_h}} \sum_{h=1}^{H} \langle q_c^{(h)},\, r_c^{(h)} \rangle. \tag{8}$$

The pooled evidence is scored along the same direction that selected it. In the pairwise mode, (6)–(8) run once per pair and give $s_{c,r}$, and the candidate keeps its evidence against its strongest rival through a soft minimum with temperature t:

$$s_c = -t\,\log\Big(\frac{1}{K-1}\sum_{r\in S\setminus\{c\}} \exp(-s_{c,r}/t)\Big), \tag{9}$$

which tends to $\min_r s_{c,r}$ as $t \to 0$ and to the mean over the rivals as $t \to \infty$ ($t = 1$ in the main study). The final logits are

$$z_c = z_c^0 + \gamma\, s_c \quad (c \in S), \qquad z_c = z_c^0 \quad (c \notin S), \tag{10}$$

with the learned scale γ initialised to 1. During training, each token of an image is masked out of the keys with probability 0.1, and at least one token is always kept.

4.5 Training objective

$$\mathcal{L} = \mathrm{CE}_{\mathrm{ls}}(z, y) + \alpha\,\mathrm{CE}_{\mathrm{ls}}(z^0, y), \tag{11}$$

where $\mathrm{CE}_{\mathrm{ls}}$ is the cross-entropy with label smoothing 0.1 and $\alpha = 1$; the second term keeps stage 1 a good shortlister. An ablation replaces the first term by a cross-entropy over the shortlist only. The same objective trains B2; B0 and B1, which have no separate stage 1, use the first term alone.

4.6 Baselines

B0, linear probe: $z = W\,\mathrm{LN}(g) + b^0$, the DINOv2 linear-evaluation recipe on the cached descriptor. *B1, attentive probe*: $M = 4$ learned class-agnostic queries attend over the tokens with the multi-head attention above (no attention bias, no score); the M pooled vectors are concatenated, layer-normalised, joined with $\mathrm{LN}(g)$ and fed to a linear classifier. It measures class-agnostic attention at a similar size (it has somewhat more parameters than the re-rankers, Table 6). *B2, class attention over all classes*: the class query $q_c = e_c$ for every one of the C classes, with (6)–(10) applied to all classes instead of a shortlist. It has exactly the parameters of every re-ranking head (Table 6).

4.7 Four facts

The following facts are elementary. We state them because they fix what the controls isolate and which variants can be told apart at all. Write $\bar{e}_S = \frac{1}{K}\sum_{j\in S} e_j$ for the mean embedding of the shortlist.

Fact 1 (Uniform rivals are shortlist centring)**.** *The rival query (2) equals* $q_c = \frac{K}{K-1}(e_c - \bar{e}_S)$. *Hence* $\sum_{c\in S} q_c = 0$, *and for* $K = 2$, $S = \{c_1, c_2\}$, *the two queries are exact negatives,* $q_{c_1} = e_{c_1} - e_{c_2} = -q_{c_2}$.

Proof. $\sum_{r\in S\setminus\{c\}} e_r = K\bar{e}_S - e_c$, so $q_c = e_c - \frac{K\bar{e}_S - e_c}{K-1} = \frac{K}{K-1}(e_c - \bar{e}_S)$. Summing over $c \in S$ gives $\frac{K}{K-1}(K\bar{e}_S - K\bar{e}_S) = 0$. For $K = 2$ the formula gives $2(e_{c_1} - \frac{1}{2}(e_{c_1} + e_{c_2})) = e_{c_1} - e_{c_2}$. □

The factor $K/(K-1)$ is absorbed by the learned embeddings, so the rival mode centres the class embeddings on the image's own shortlist. The class, centre, random and rival queries therefore lie on one axis: no centring, centring on all classes, centring on a random set, centring on the shortlist. A difference between rival and class alone says nothing about shortlist conditioning; differences between rival and both centre and random do (Table 8).

Fact 2 (Belief-weighted rivals with tied weights are a log-softmax gradient)**.** *Suppose the class embeddings were the weights* w_j *of a linear classifier acting on a pooled feature* f, *and let* $p_j = \exp(w_j^\top f)/\sum_{j'\in S}\exp(w_{j'}^\top f)$ *be the softmax over* S. *Then* $\nabla_f \log p_c = (1-p_c)(w_c - \sum_{r\in S\setminus\{c\}} \pi_r\, w_r)$ *with* $\pi_r = p_r/(1-p_c)$, *and* π_r *is the softmax over the rivals of the logits* $w_r^\top f$.

Proof. $\nabla_f \log p_c = w_c - \sum_{j\in S} p_j w_j = (1-p_c)\, w_c - \sum_{r\neq c} p_r w_r$. Since $1 - p_c = \sum_{r\neq c} p_r$, dividing the last sum by $1 - p_c$ gives weights π_r that sum to one and are proportional to $\exp(w_r^\top f)$. □

With tied weights, identity projections and $T = 1$, the rival-soft query would thus be the shortlist-restricted log-softmax gradient up to the factor $1 - p_c$, and its attention logits a gradient map of the log-softmax with SGLRP's weighting [9], computable after training without new parameters. This is why e_c is untied from W, why W_k and W_v are learned, and why the weights use the detached z^0: the second stage cannot reshape the first stage's probabilities to suit itself.

Fact 3 (The score is a soft maximum of a per-patch margin)**.** *Take one head,* $d = D$, *identity projections without layer normalisation (*$k_i = v_i = x_i$*) and* $b = 0$, *and let* $m_i = \langle q_c, x_i\rangle/\sqrt{d}$. *Then* $s_c = \sum_i \mathrm{softmax}_i(m_i/\tau)\, m_i$. *This value lies between* $\frac{1}{N}\sum_i m_i$ *and* $\max_i m_i$; *it tends to* $\max_i m_i$ *as* $\tau \to 0$ *and to the mean as* $\tau \to \infty$. *For the rival query,* m_i *is proportional to* $\langle e_c - \bar{e}_S, x_i\rangle$, *the margin of patch* i *for* c *over the shortlist's mean embedding.*

Proof. Substituting into (6)–(8) gives the expression. Its weights sum to one and increase with m_i, so the weighted mean is at least the plain mean (Chebyshev's sum inequality) and at most the maximum; the limits are those of the softmax. The rival case follows from Fact 1. □

This is CSRA's average-plus-max form [3] with the class weight replaced by the contrastive query.

Fact 4 (Pairwise equals rival at $K = 2$ and is not a single query for $K \geq 3$)**.** *(a) For* $K = 2$ *the pairwise and rival modes compute the same function with the same parameters. (b) For* $K \geq 3$, *distinct embeddings and generic parameters, no single query per candidate, centred or not, reproduces the pairwise score on all inputs.*

Proof. (a) With $S = \{c, r\}$, $q_{c,r} = e_c - e_r$ is the rival query (Fact 1), and (9) over a single rival returns $s_{c,r}$. The repository's test `test_pairwise_with_two_candidates_equals_rival` asserts this numerically. (b) Consider images whose N tokens are all equal, $x_i = x$, with g, and hence S, held fixed. Every attention map is then uniform and $r^{(h)} = v^{(h)}$ with $v = W_v\,\mathrm{LN}(x)$, so a single query q built from the embeddings scores $\langle q, v\rangle/\sqrt{d_h}$. Let x run along a great circle of the

sphere on which the normalised part of LN lives; then $v(\theta) = a + b\cos\theta + c\sin\theta$ for fixed vectors a, b, c, and every single-query score has the form $\alpha + \beta\cos\theta + \delta\sin\theta$. The pairwise score is the soft minimum (9) of $K-1 \geq 2$ functions of this form, one per rival. Take two rivals r_1, r_2 with scores u_1, u_2 and write $\Delta = u_2 - u_1$; with only these two, the soft minimum equals $u_1 - t\log(\frac{1}{2}(1+e^{-\Delta/t}))$. The last term is a strictly concave function of Δ, and $\Delta(\theta) = \Delta_0 + \rho\cos(\theta - \varphi)$ is not constant whenever $e_{r_1} - e_{r_2}$ is not orthogonal to both b and c, which holds for generic parameters. A strictly concave function of $\Delta_0 + \rho\cos\psi$ with $\rho \neq 0$ is not of the form $\alpha + \beta\cos\psi + \delta\sin\psi$ (being even in ψ, it would have to be affine in $\cos\psi$ on $[-1, 1]$), so for $K = 3$ no single query reproduces the score. With more than two rivals the same holds for generic parameters. □

The pairwise mode is the learned, visual counterpart of FuDD's pairwise descriptions [7] and of POOF's pair features [64], formed per image. Hypothesis H4 asks whether it adds anything beyond centring at $K = 5$.

4.8 Complexity

Table 6 gives the parameter counts (as the design document records them from the code's parameter counter; the formulas reproduce them) and the multiply-accumulate operations (MACs) of the head per image, ignoring layer normalisations and softmaxes. All re-ranking modes, including pairwise, have exactly the parameters of B2, so every comparison among them and with B2 is at equal capacity. With $K = 5$ the per-query part of a re-ranker costs 1/26 of B2's ($K = 5$ against $C = 130$ queries); since keys and values are shared, the whole head costs about half (17.6 M against 34.0 M MACs). The attention tensor has $B \cdot H \cdot Q \cdot N$ entries for Q queries per image, so memory is linear in the number of queries and the pairwise mode multiplies it by $K - 1$. For scale, the frozen ViT-B/14 at 224 px costs about 23.5 G MACs ($12 \times (12\,n \cdot 768^2 + 2\,n^2 \cdot 768)$ with $n = 261$ tokens, plus the patch embedding), so a re-ranking head adds under 0.1 % to inference. Projecting the 256 tokens on 128 principal components adds $256 \cdot 768 \cdot 128$, about 25.2 M MACs (our count). Measured FLOPs, latency and memory are results (Section 6.10).

Table 6: Parameters and multiply-accumulates per image of the heads ($D_g = 1536$, $D = 128$, $d = 256$, $H = 4$, $N = 256$). MACs are for Tsinghua Dogs ($C = 130$) and leave out layer normalisations and softmaxes.

Head	Parameters	Tsinghua ($C = 130$)	Stanford ($C = 120$)
B0, linear probe	$2D_g + (D_g + 1)C$	202,882	187,512
B1, attentive probe ($M = 4$)	$2D + 2Dd + Md + 2D_g + 2Md + (D_g + Md + 1)C$	404,866	379,256
B2 and every re-ranking mode	B0 $+ Cd + 2D + 2Dd + d + 1$	302,211	284,281
Part of the head	**MACs**		**per image ($C = 130$)**
keys and values, shared	$2NDd$		16,777,216
global head	D_gC		199,680
one query: attention, pooling, score	$2Nd + d$		131,328
re-ranking, $K = 5$ queries	$K(2Nd + d)$		656,640
pairwise, $K(K-1) = 20$ queries	$K(K-1)(2Nd + d)$		2,626,560
B2, $C = 130$ queries	$C(2Nd + d)$		17,072,640
total: re-ranking / pairwise / B2			17,633,536 / 19,603,456 / 34,049,536

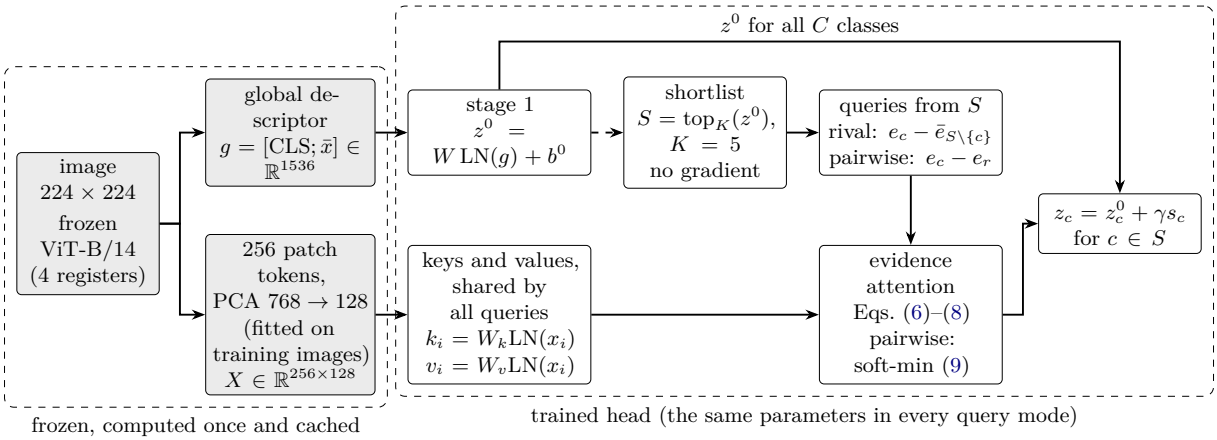


Figure 1: RCEA. The frozen backbone runs once per image and its outputs are cached (grey). The trained head has a linear stage 1 on g and a second stage on the tokens: the shortlist is taken without gradient (dashed arrow), each candidate's query is built from the shortlist, attends over the tokens and gives a score that is added to that candidate's stage-1 logit; logits outside the shortlist pass through unchanged. The controls replace only the box "queries from S".

Algorithm 1 One forward pass of the re-ranking head on a batch (as implemented in `CandidateReRanker.forward`).

	input: $g \in \mathbb{R}^{B\times D_g}$, $X \in \mathbb{R}^{B\times N\times D}$, labels y (training only)	
1	$z^0 \leftarrow W\,\mathrm{LN}(g) + b^0$	*stage-1 logits, $B \times C$*
2	$S \leftarrow \mathrm{top}_K\big(\mathrm{sg}(z^0)\big)$	*indices, $B \times K$, no gradient*
3	**if** training **and** `gt_insert`: for each image with $y \notin S$, put y in the last slot of S	*the truth replaces the weakest candidate*
4	$E \leftarrow (e_c)_{c\in S}$	$B \times K \times d$
5	**if** mode is pairwise: $Q \leftarrow (e_c - e_r)_{c\in S,\, r\in S\setminus\{c\}}$	$B \times K(K-1) \times d$
6	**else**: $Q \leftarrow \mathrm{query}(E, \mathrm{mode})$	*Eqs. (2), (3) or a control (5)*
7	$k_i \leftarrow W_k\,\mathrm{LN}(x_i)$, $v_i \leftarrow W_v\,\mathrm{LN}(x_i)$, split into H heads	*once per image, shared by all queries*
8	$A \leftarrow \mathrm{softmax}_i\big((Q+b)^\top k_i/(\sqrt{d_h}\,\tau)\big)$ per head	*training: each key masked with probability 0.1*
9	$R \leftarrow A\,V$	$B \times H \times Q \times d_h$
10	$s \leftarrow \sum_h \langle Q^{(h)}, R^{(h)}\rangle/\sqrt{d_h}$	*one score per query*
11	**if** mode is pairwise: $s_c \leftarrow$ soft-min$_t$ of $(s_{c,r})_{r\neq c}$	*Eq. (9)*
12	$z \leftarrow z^0$; $z_c \leftarrow z_c + \gamma\, s_c$ for $c \in S$	*only the K candidates move*
13	**return** z; in training the loss $\mathrm{CE}_{\mathrm{ls}}(z, y) + \alpha\,\mathrm{CE}_{\mathrm{ls}}(z^0, y)$	*Eq. (11)*

5 Experimental protocol

5.1 Features

The main backbone is DINOv2 ViT-B/14 with four registers (timm `vit_base_patch14_reg4_dinov2.lvd142m`), frozen, at 224 px. The evaluation transform resizes the shorter side to 256 pixels (bicubic), takes the centre 224×224 crop and applies the ImageNet normalisation. The cache holds $g = [\mathrm{CLS}; \bar{x}]$ (1536 dimensions) and the 256 patch tokens projected on a 256-component PCA. The PCA is fitted on the tokens of 4,000 randomly chosen training images (seed 0) of each dataset; Stanford Dogs has its own PCA. The 256 components keep 84.2 % of the token variance on Tsinghua Dogs and 83.3 % on Stanford Dogs. The heads read the first 128 components (32 or 64 in the token-width study). The first 32, 64 and 128 components keep 46.4, 58.8 and 71.7 % of the token variance on Tsinghua Dogs, and 43.3, 56.3 and 70.0 % on Stanford Dogs (read from the saved PCA bases, each fitted on 1,024,000 tokens). The cached views are the original images of all three splits, the horizontally flipped training images, and the

test views of Section 5.8; for the transfer study the Stanford test images are also cached in the Tsinghua PCA basis. Tokens are stored as float16. The backbone runs in fp16 autocast on the T4; the heads train and predict in fp32. On Tsinghua Dogs the extraction ran at about 90–140 images per second, limited by the two CPU cores decoding JPEGs, with a peak GPU memory of 1.20 GB and a cache of 24 GB. The training tokens (original and flipped, 7.7 GB) are kept on the GPU during training.

5.2 Heads and training

Nine heads are trained on each main dataset: the baselines B0, B1 and B2; the controls class, centre and random; and the proposed rival, rival-soft and pairwise, all re-rankers with $K = 5$ (Section 4). All heads use the same optimiser, schedule, epochs and features (Appendix D): AdamW with weight decay 0.05 on weight matrices and none on vectors; a linear warm-up over one epoch followed by cosine decay over at most 30 epochs; batch size 256; gradient clipping at norm 1; label smoothing 0.1; token dropout with probability 0.1 for every head that reads tokens (B1, B2 and all re-rankers); no dropout. Each training image uses its cached original or flipped view, chosen at random. Early stopping keeps the epoch with the highest validation top-1 (ties: lower validation NLL) and stops after 8 epochs without a new best. The test split is not loaded by any training script.

5.3 Model selection, seeds and the freeze

For each head, the learning rate is chosen from $\{3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}\}$ on seed 0 by validation top-1 (ties: lower validation NLL). Seeds 0, 1 and 2 are then trained at that rate, so every head has three seed runs. The ablation, open-set and token-width runs borrow the rate of the matching head of the Tsinghua main experiment and select nothing themselves; each other backbone runs its own selection over the same grid, since one rate does not carry over between feature spaces. In both main experiments every head chose $3 \cdot 10^{-4}$, the lowest rate in the grid (recorded in the freezes of 20:13:39 and 21:03:42 UTC), and so did every head of the three other backbones. The best rate may lie below the grid; the grid was fixed before training and was not widened afterwards, which is a limitation (Section 7).

The selection is frozen in code. `scripts/freeze.py` writes, per experiment, the time, each head's rate and, for every seed run, its validation metrics, best epoch and the SHA-256 of its checkpoint. Every evaluation script refuses to run on an experiment without a freeze, on a run the freeze does not list, or on a checkpoint whose hash has changed. The Stanford main experiment was frozen at 20:13:39 UTC and the Tsinghua main experiment at 21:03:42 UTC on 8 November 2026, and both freeze summaries were entered in the research log before their test evaluations (Appendix E). The seven secondary experiments are finished by a watcher (`scripts/auto_final.sh`, declared in the log at 21:42 UTC, before any of them had finished), which freezes and then evaluates each one as soon as its training ends. Their freeze summaries are copied into the log afterwards, from `freeze.json`, whose time stamp and the first time stamp of the evaluation log show the order. This departs from the stated order (freeze recorded in the log before the evaluation); no choice is made between the end of training and the freeze. Only one of the seven finished before the free-tier GPU runtime ended at about 22:14 UTC, after which Colab's usage limit refused a new one: `tsinghua_resnet50`, frozen at 21:56:00 UTC (12 runs, 4 heads) and evaluated from 21:56:02 to 21:57:39 UTC, with its freeze summary entered in the log afterwards as described. `tsinghua_ablate` and `tsinghua_convnext_tiny` were cut off during training, and `tsinghua_openset`, `tsinghua_clip_vitb16`, `tsinghua_tok32` and `tsinghua_tok64` never started. None of these six was frozen, and no test split was read for any of them.

Resumption on Kaggle. The six were finished on 9 November in a free-tier Kaggle session with two T4 GPUs (`scripts/kaggle_run.sh`), from the repository and the work folder mirrored on Google Drive (runs with their checkpoints, logs, manifests and the saved token PCA basis), both packed as archives and checked by SHA-256 on arrival. The feature caches had not been saved, so they were rebuilt from the three Tsinghua archives, downloaded again and matched against the hashes of the Colab copies (Appendix B). The software differs (torch 2.11.0, cuDNN 92700, on Kaggle; torch 2.14.1, cuDNN 92800, on Colab), so the forward pass is not bit-identical. A resume rule, recorded in the research log at 23:35 UTC on 8 November before any check had run, decided whether a cut-off experiment keeps its finished runs: only if none of them, applied to the rebuilt validation features, moves its validation top-1 by more than 0.001 (about 7 of 6,515 images); otherwise the whole experiment is trained again, so that all its runs read the same features (`scripts/feature_drift_check.py`, `scripts/resume_policy.py`; no test split is read). The frozen heads of `tsinghua_main` were checked the same way, for the record. The largest change was 0.0003 for the 24 deterministic heads of `tsinghua_main` (at most 3 of the 6,515 validation predictions changed) and for the 22 finished ablation runs, and zero for the 6 finished ConvNeXt-T runs, so both experiments kept their finished runs and trained only the missing ones. As declared, the random-rival control is not part of this check, because its predictions depend on its random draws (59 to 82 of its validation predictions changed). A watcher froze each experiment when its training ended and then evaluated it, as on Colab; both new backbones chose the learning rate $3 \cdot 10^{-4}$ again for every head. The freezes (UTC) were `tsinghua_convnext_tiny` at 02:40:25 (12 seed runs, 4 heads), `tsinghua_ablate` at 03:06:03 (39 runs, 13 heads), `tsinghua_clip_vitb16` at 03:08:16 (12, 4), `tsinghua_openset` at 03:23:51 (12, 4), `tsinghua_tok32` at 03:38:12 (9, 3) and `tsinghua_tok64` at 03:52:42 (9, 3). Each test evaluation began within a second of its freeze and took 18 to 69 seconds, and the freeze summaries were entered in the research log afterwards, as for `tsinghua_resnet50`.

5.4 Hypotheses and the decision rule

The hypotheses (Table 7) and the decision rule were written down before the first training run and made mechanical by seven analysis rules recorded before any test evaluation (Appendix F). The primary metric is test top-1, and Tsinghua Dogs is the primary dataset. For a comparison of head A with head B on seed s, let Δ_s be the difference of their test top-1 on the same test images and $[\ell_s, u_s]$ its paired bootstrap 95 % interval. The rule reads

$$\text{“}A > B\text{”} \iff \Delta_s > 0 \text{ for all } s \in \{0,1,2\} \text{ and } \ell_s > 0 \text{ for at least two } s, \tag{12}$$

symmetrically for “$B > A$” (all $\Delta_s < 0$ and $u_s < 0$ for at least two seeds); otherwise no claim is made, and a comparison with fewer than three seeds is incomplete. The exact McNemar test is reported beside it but is not part of the rule. The rule is applied to the numbers as they come out, and a refuted hypothesis is reported as refuted.

How the hypotheses are evaluated. The table code computes every verdict from the digest. *H1*: the best proposed mode and the best control are the tags with the highest mean validation top-1 over their three seed runs on Tsinghua Dogs (ties: lower mean validation NLL); Stanford Dogs repeats the same pair, and its own validation picks are reported beside it. H1 is supported if the pair gives “$A > B$” on Tsinghua and a positive mean difference on Stanford. Applied to the frozen validation predictions of the Tsinghua main experiment, this rule names rival-soft as the best proposed mode and the class query as the best control; H2, H5 and H7 use rival-soft. *H2*: a fix is an image whose stage-1 top-1 is wrong and whose final top-1 is right, a break the reverse. The code compares the two rates over all test images, which compares their numbers; the design's restriction to images with the truth in S changes nothing for breaks, and for fixes only the rare case in which all K candidates are pushed below a truth outside S. The hard

Table 7: The seven hypotheses as fixed before training (design document, Sec. 8); names shortened, predictions and refutation criteria as there. "The rule" is Eq. (12).

	Hypothesis	Prediction	Refuted if
H1	Shortlist conditioning matters	rival, rival-soft or pairwise > class, centre and random on the Tsinghua test split; the same sign on the Stanford test split	the best proposed mode does not exceed the best control by the rule
H2	It acts where it should	among images whose truth is in S but not top-1 under z^0, fixes far outnumber breaks; the gain over B0 concentrates in the 20 breed pairs most confused by B0	fix rate $\leq$ break rate, or the gain on the 20 pairs is no larger than the overall gain
H3	It is the confusion set, not contrast in general	random $\leq$ centre	random within the seed spread of rival
H4	Pairwise evidence adds beyond centring	pairwise $\geq$ rival at $K = 5$ (identical at $K = 2$, Fact 4)	exploratory; reported either way
H5	The shortlist costs nothing	re-ranking $\geq$ B2 at 1/26 of its per-query cost	B2 is ahead by the rule
H6	K behaves like a shortlist	over $K \in \{2, 3, 5, 10\}$ the shortlist recall rises with K and accuracy is flat or peaks at a small K	accuracy falls monotonically from $K = 2$
H7	Robustness (exploratory)	under corruptions that lower top-1 but keep the top-K recall, the re-ranker recovers more than B0	reported either way

pairs are the 20 unordered breed pairs with most confusions in either direction in the validation predictions of B0, seed 0. *H3*: "within the seed spread" means |random − rival| is at most the larger of their two test top-1 standard deviations over seeds; if not, H3 is supported when random $\leq$ centre and partly supported otherwise. *H5*: refuted if B2 is ahead by the rule; supported with "re-ranker ahead" or "no difference" otherwise. *H6*: uses the rival head; refuted if top-1 falls strictly from $K = 2$ to 3, 5 and 10. *H4* and *H7* are exploratory and reported as the rule's outcome and as the per-view gains over B0 against the clean gain.

5.5 Comparisons

The pairs are fixed in `scripts/final_all.sh` together with the views and their order. On the clean test view of both main experiments there are 22 pairs: each proposed head against each of the three controls (9 pairs); pairwise and rival-soft against rival; the control chain random against centre and centre against class; each proposed head against B2 (3); each proposed head and the class query against B0 (4); rival and pairwise against B1. Table 8 lists what the main comparisons isolate. On every corrupted or annotation-assisted view, 7 pairs are compared: rival, rival-soft, pairwise and class against B0, and rival, rival-soft and pairwise against class. The ablation compares rival with class at $K = 2, 3, 10$ and pairwise with rival at $K = 3, 10$; each other backbone compares rival and pairwise with class, pairwise with rival, and rival and class with B0; the token-width runs compare rival with class and rival and class with B1. The provenance subsets repeat the 22 clean-view pairs (Section 5.11).

5.6 Statistics

Every head has three seed runs. Summaries are the mean and standard deviation over seeds (with $n - 1$ in the denominator) and a t-based 95 % interval. Paired comparisons are made seed by seed on the same test images: a percentile bootstrap of the top-1 difference with 10,000 resamples of the test images (generator seed 0), and the exact McNemar test on the discordant

Table 8: What each main comparison isolates. All pairs are evaluated on the clean test view of both main experiments.

Comparison	Isolates
class (re-ranking) vs B0	a second look at the patches with class-specific queries (CSRA on the shortlist)
centre vs class	centring alone (global, the same for every image)
random vs centre	contrast against a random set of classes
rival vs random	contrast against the image's own confusion set (the claim)
rival-soft vs rival	weighting the rivals by the stage-1 belief
pairwise vs rival	pairwise evidence against shortlist centring ($K \geq 3$)
re-ranking vs B2	shortlist against all classes, at equal parameters
re-ranking vs B1	class-conditional against class-agnostic attention, at similar capacity

images (a two-sided binomial test of n_{10} out of $n_{10} + n_{01}$ with probability 1/2). The verdict is Eq. (12). For the random control, the standard deviation of its test top-1 over five further draws of the random classes (generator seeds $s, 1000 + s, \ldots, 4000 + s$ for seed run s) is reported as `top1_draw_sd`. Appendix G gives the details.

5.7 Metrics

Top-1 and top-5 accuracy; macro precision, recall and F1; NLL, expected calibration error (15 equal-width bins) and Brier score, raw and after temperature scaling, with the temperature fitted on the run's validation predictions [90, 91]; the area under the risk–coverage curve and the risk at 90 % coverage of the maximum softmax probability [92]. For re-ranking heads also: the stage-1 top-1, the shortlist recall (truth in S), and the shares of all test images whose top-1 the second stage changes, fixes (stage 1 wrong, final right) or breaks (stage 1 right, final wrong). Per breed: the test accuracy of each head. For the hard pairs: the test errors within the 20 pairs and the accuracy on the test images of their breeds.

5.8 Test views

Each view is applied to the full image before the evaluation transform, deterministically, with the image id as seed (Table 9). The automatic views use no annotation. The annotation-assisted views use the dataset's ground-truth boxes at test time. They are robustness probes and are never reported as automatic inference; their results are kept in a separate block of every table and figure. Stanford Dogs has body boxes only, so it has three annotation-assisted views. H7 uses the automatic views only.

5.9 Open set with held-out breeds

30 of the 130 Tsinghua breeds are drawn at random (seed 0) and held out; the draw is stored in `configs/openset/tsinghua_unknown30_s0.json`. B0 and the class, rival and pairwise re-rankers are trained on the remaining 100 breeds, with no image of a held-out breed in training or validation, at the rates of the main experiment. On the test split, the images of the 100 known breeds are in-distribution and those of the 30 held-out breeds are unknown. Scores (larger means known): maximum softmax probability, raw and temperature-scaled; maximum logit; energy (log-sum-exp of the logits); and the top-1 minus top-2 logit margin; for re-rankers also the same scores of the stage-1 logits. Metrics: AUROC and the false-positive rate at 95 % true-positive rate (threshold at the 5 % quantile of the known images' scores), and closed-set top-1 on the known breeds. Two caveats hold by construction: the held-out breeds are unknown to the heads but not to the backbone, and the token PCA is the main one, fitted without labels on training images of all 130 breeds.

Table 9: Test views. Grey is (124, 116, 104), the ImageNet mean. An annotation-assisted view leaves an image unchanged when the image has no usable box.

View	What it does	Boxes at test time	Datasets
clean	none (the evaluation transform only)	no	both
blur	Gaussian blur, radius 3	no	both
noise	additive Gaussian noise, $\sigma = 25$ on the 0–255 scale	no	both
JPEG	re-encoding at JPEG quality 10	no	both
low resolution	bilinear down-sampling by 4 and back up	no	both
occlusion	a grey square covering 25 % of the image at a seeded position	no	both
head occlusion	the head box painted grey	yes	Tsinghua
grey background	everything outside the body box painted grey	yes	both
blurred background	everything outside the body box blurred, radius 12	yes	both
body crop	crop to the body box with a 10 % margin	yes	both
head crop	crop to the head box with a 10 % margin	yes	Tsinghua

5.10 Transfer to de-duplicated Stanford Dogs

Heads trained on Tsinghua Dogs label Stanford Dogs test images without any training on Stanford Dogs. The breed correspondence is the one the audit measured: for each Tsinghua breed, the majority Stanford breed among its confirmed pairs. A Stanford breed is used only if exactly one Tsinghua breed maps to it, with a purity of at least 0.9 and at least 5 pairs. The images are the 973 clean Stanford test images (no candidate in the Tsinghua training or validation data) of the mapped breeds, with tokens in the Tsinghua PCA basis. The prediction is the full 130-breed Tsinghua decision and counts as correct when it is the mapped breed; a second number restricts the decision to the mapped breeds. These images are still ILSVRC-2012 training images (Section 3.5).

5.11 Provenance subsets

`scripts/subset_eval.py` re-scores the saved test predictions of the Tsinghua main experiment on the subsets of Section 3.4 (all, Stanford-origin, independent) and repeats the 22 clean-view comparisons on each. Nothing is trained or selected on the subsets, and they were fixed by the audit before any test prediction existed.

5.12 Other backbones

Three further frozen backbones are run on Tsinghua Dogs with B0 and the class, rival and pairwise re-rankers: CLIP ViT-B/16 (timm `vit_base_patch16_clip_224.openai`, WIT-400M; 196 tokens; $g = [\mathrm{CLS}; \bar{x}]$ of 1536 dimensions; CLIP's own normalisation), ResNet-50 (torchvision `IMAGENET1K_V2`; 49 tokens of 2048 dimensions; g the average-pooled feature) and ConvNeXt-T (torchvision `IMAGENET1K_V1`; 49 tokens of 768 dimensions; g the average). Each has its own 256-component token PCA (first 128 components used), its own learning-rate grid, and only the clean and flipped views; it is evaluated on the clean test view only.

5.13 Efficiency

`scripts/efficiency_report.py` measures, on the runtime's device (whose name it records) and with nothing else on the GPU: parameters; FLOPs from `torch.utils.flop_counter`, where a multiply-add counts as 2 FLOPs; median latency over 50 timed passes after 10 warm-up passes; and peak GPU memory. It covers every head on random inputs of the cached shapes at batch 1

and 256 (fp32), the frozen backbone at batch 1 and 64 (fp16 autocast), and the full single-image pipeline (backbone, token PCA, head). It reads no test data and is not part of the locked evaluation.

5.14 Attention against the boxes

`scripts/explain.py` records, for seed 0 of each head on the Tsinghua test split, the attention over the token grid, averaged over attention heads (for pairwise, also over the candidate's rival pairs): for re-rankers the map of the final top-1 candidate, for B1 the mean of its queries, for B2 the map of the predicted class. Using the dataset's head and body boxes for this measurement only, it reports the attention mass inside each box, the box's share of the image (what a uniform map would give), their ratio (the lift; above 1 means the map prefers the box) and the entropy of the map relative to a uniform one. Its figure is chosen without hand-picking: the first images by id of each kind (the rival head fixed a stage-1 error, broke a correct stage-1 answer, or both were right), with the maps of the two highest-scoring final candidates of the rival and class heads.

6 Results

Each table and figure is produced from the result digest (`results/digest`, 15 files whose SHA-256 are listed in its `MANIFEST.txt`) by `scripts/make_tables.py` and `scripts/make_figures.py`. The secondary studies were finished on Kaggle and merged into the digest with `scripts/ingest_kaggle.sh`, which checks the result archives against their SHA-256 and requires the archive's run registry to contain every row of the earlier one. The numbers in the text are read from the same tables and from `paper/generated/numbers.json`. Differences are in percentage points (pp) of test top-1 unless stated otherwise, and "mean" is the mean over the three seed runs.

6.1 Main results

On Tsinghua Dogs the three baselines are level: 86.26 % (B0), 86.26 % (B1) and 86.28 % (B2) test top-1 (Table 10). Every re-ranking head is above them, from 86.43 % for random rivals to 87.12 % for rival-soft, the highest mean (standard deviation 0.15 over seeds); top-5 lies between 98.45 and 98.58 % for all nine heads, and macro-F1 follows top-1 (86.23 % for B0, 87.00 % for rival-soft). On Stanford Dogs the picture differs (Table 11): B0 (88.33 %), B2 (88.34 %) and the six re-rankers (88.16–88.28 %) lie within 0.2 points of each other, and the attentive probe trails at 86.71 %. Which of these differences the rule supports is stated in Section 6.2 and Table 13.

After temperature scaling every head is well calibrated, with ECE between 1.16 and 1.61 % on Tsinghua Dogs and between 1.16 and 1.49 % on Stanford Dogs. On Tsinghua Dogs the re-rankers have a lower NLL (0.461–0.472 against 0.479–0.482 for the baselines) and Brier score (0.217–0.225 against 0.231–0.233), rival-soft the lowest of both, and selective prediction follows: the area under the risk–coverage curve (AURC) is 3.36 % for rival-soft against 3.74 % for B0, and the error rate on the 90 % most confident test images is 8.74 % against 9.46 %. On Stanford Dogs B0 has the lowest NLL (0.402; the others 0.406–0.413, B1 0.453), and the AURC is 2.76–2.85 % for every head except B1 (3.53 %). No test was pre-registered for these metrics; they are descriptive.

6.2 The hypotheses under the decision rule

Table 12 gives the verdicts as computed. Each paragraph below states the verdict first and then the evidence; a refuted hypothesis is reported as refuted.

Table 10: Tsinghua Dogs test split (5,200 images): the nine heads on the same frozen DINOv2 features. Mean and standard deviation over three seeds of top-1, top-5, macro precision, recall and F1, and of NLL and ECE after temperature scaling fitted on validation. Bold marks the highest mean top-1; differences between heads are judged only by the paired comparisons of Table 13.

Head	Params	Top-1	Top-5	Macro-P	Macro-R	Macro-F1	NLL_{TS}	ECE_{TS} (%)
Baselines								
Linear probe (B0)	202,882	86.26 ± 0.21	98.49 ± 0.03	88.19 ± 0.10	86.26 ± 0.21	86.23 ± 0.28	0.480 ± 0.009	1.26 ± 0.15
Attentive probe (B1)	404,866	86.26 ± 0.02	98.58 ± 0.09	88.20 ± 0.05	86.26 ± 0.02	86.33 ± 0.05	0.479 ± 0.004	1.61 ± 0.23
Class attention, all classes (B2)	302,211	86.28 ± 0.29	98.53 ± 0.16	88.25 ± 0.15	86.28 ± 0.29	86.26 ± 0.32	0.482 ± 0.008	1.48 ± 0.10
Equal-size controls								
Re-rank: class query	302,211	86.71 ± 0.10	98.56 ± 0.09	88.39 ± 0.03	86.71 ± 0.10	86.71 ± 0.10	0.466 ± 0.001	1.32 ± 0.30
Re-rank: centred query	302,211	86.49 ± 0.17	98.45 ± 0.04	88.22 ± 0.11	86.49 ± 0.17	86.45 ± 0.24	0.470 ± 0.008	1.27 ± 0.04
Re-rank: random rivals	302,211	86.43 ± 0.21	98.55 ± 0.10	88.28 ± 0.09	86.43 ± 0.21	86.41 ± 0.29	0.472 ± 0.011	1.47 ± 0.29
Proposed (RCEA)								
RCEA: rival	302,211	86.74 ± 0.23	98.55 ± 0.08	88.36 ± 0.13	86.74 ± 0.23	86.66 ± 0.27	0.467 ± 0.009	1.45 ± 0.14
RCEA: rival-soft	302,211	**87.12 ± 0.15**	98.53 ± 0.05	88.52 ± 0.19	87.12 ± 0.15	87.00 ± 0.24	0.461 ± 0.009	1.40 ± 0.53
RCEA: pairwise	302,211	86.80 ± 0.12	98.46 ± 0.04	88.34 ± 0.09	86.80 ± 0.12	86.76 ± 0.17	0.463 ± 0.003	1.16 ± 0.27

Table 11: Stanford Dogs test split (8,580 images), read as Table 10. Absolute values are optimistic for every head, since every Stanford Dogs image is an ILSVRC-2012 training image (Section 3.5).

Head	Params	Top-1	Top-5	Macro-P	Macro-R	Macro-F1	NLL_{TS}	ECE_{TS} (%)
Baselines								
Linear probe (B0)	187,512	88.33 ± 0.07	98.82 ± 0.10	88.05 ± 0.07	87.98 ± 0.11	87.84 ± 0.11	0.402 ± 0.004	1.49 ± 0.29
Attentive probe (B1)	379,256	86.71 ± 0.49	98.66 ± 0.11	86.59 ± 0.32	86.43 ± 0.40	86.22 ± 0.44	0.453 ± 0.011	1.45 ± 0.34
Class attention, all classes (B2)	284,281	**88.34 ± 0.08**	98.77 ± 0.05	88.09 ± 0.06	87.97 ± 0.02	87.84 ± 0.03	0.410 ± 0.004	1.22 ± 0.19
Equal-size controls								
Re-rank: class query	284,281	88.28 ± 0.05	98.78 ± 0.03	88.03 ± 0.08	87.92 ± 0.06	87.79 ± 0.04	0.410 ± 0.004	1.16 ± 0.21
Re-rank: centred query	284,281	88.26 ± 0.09	98.81 ± 0.10	88.02 ± 0.12	87.89 ± 0.12	87.76 ± 0.13	0.406 ± 0.007	1.35 ± 0.31
Re-rank: random rivals	284,281	88.16 ± 0.10	98.72 ± 0.05	87.91 ± 0.13	87.87 ± 0.14	87.68 ± 0.10	0.413 ± 0.001	1.21 ± 0.06
Proposed (RCEA)								
RCEA: rival	284,281	88.17 ± 0.12	98.79 ± 0.13	87.90 ± 0.19	87.81 ± 0.13	87.64 ± 0.12	0.408 ± 0.008	1.35 ± 0.24
RCEA: rival-soft	284,281	88.25 ± 0.19	98.78 ± 0.12	88.00 ± 0.27	87.92 ± 0.17	87.76 ± 0.19	0.409 ± 0.007	1.26 ± 0.27
RCEA: pairwise	284,281	88.18 ± 0.12	98.79 ± 0.13	87.91 ± 0.19	87.82 ± 0.12	87.65 ± 0.11	0.408 ± 0.008	1.39 ± 0.22

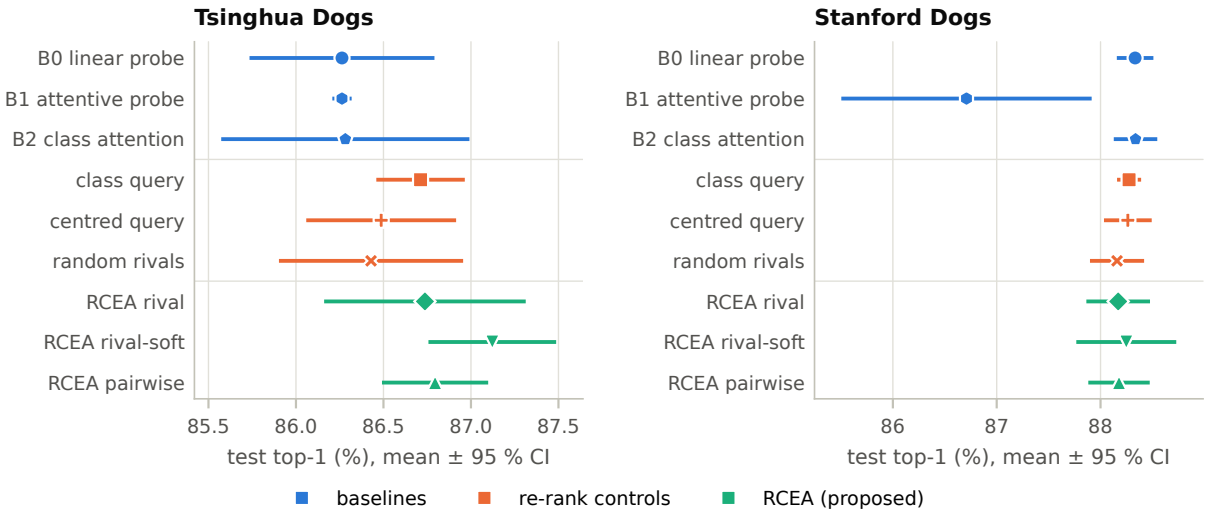


Figure 2: Test top-1 of every head, mean and t-based 95 % interval over three seeds, one panel per dataset; the colour gives the family (baselines, controls, proposed). The intervals describe seed-to-seed spread; differences between heads are judged by the paired tests of Figure 3, not by whether these intervals overlap.

Table 12: H1–H7 as judged by the pre-registered rule (Eq. (12) and Section 5.4): verdict and evidence, computed by `make_tables.py` from the digest. Tsinghua Dogs is primary; Stanford Dogs gives the sign check of H1. The table's "§8" is Section 8 of the design document, i.e. the rule of Eq. (12).

	Hypothesis	Verdict	Evidence (test split; decision rule of §8)
H1	Shortlist conditioning beats equal-size controls	refuted	Tsinghua: rival-soft vs class (chosen on Tsinghua val): n.s., per seed +0.37/+0.50/+0.37 pp; Stanford-origin n.s.; independent n.s.; Stanford: rival-soft vs class (chosen on Tsinghua val): n.s., per seed +0.05/+0.15/-0.28 pp; own val pick pairwise vs random: n.s.
H2	It acts where it should	supported	Tsinghua: fix 1.91% vs break 0.68%; gain on pairs +1.34 vs overall +0.86 pp; Stanford: fix 0.42% vs break 0.42%; gain on pairs -0.12 vs overall -0.08 pp
H3	It is the confusion set, not contrast	supported	Tsinghua: random 86.43, centre 86.49, rival 86.74 (spread 0.23); Stanford: random 88.16, centre 88.26, rival 88.17 (spread 0.12)
H4	Pairwise adds beyond centring	exploratory: no difference	Tsinghua: n.s. (+0.06 pp); Stanford: n.s. (+0.01 pp)
H5	The shortlist costs nothing vs all classes	supported (reranker better)	Tsinghua: A>B (+0.84 pp); Stanford: n.s. (-0.09 pp)
H6	K behaves like a shortlist	supported	K=2: 86.85; K=3: 86.91; K=5: 86.74; K=10: 86.52 (recall 94.2, 96.8, 98.5, 99.4)
H7	Robustness under corruptions	exploratory	Tsinghua (rival-soft − B0, pp): clean +0.9, blur +3.0, noise +0.8, JPEG +0.8, low-res +1.7, occlude +2.6; Stanford (rival-soft − B0, pp): clean -0.1, blur -0.2, noise -0.1, JPEG -0.2, low-res -0.1, occlude -0.2

H1, shortlist conditioning: refuted. Validation names rival-soft as the best proposed mode and the class query as the best control. On the Tsinghua test split rival-soft is ahead in every seed, by +0.37, +0.50 and +0.37 points, but only the second seed's paired interval excludes zero ($[+0.10, +0.90]$; the other two are $[-0.12, +0.85]$ and $[-0.10, +0.83]$), so the rule makes no claim; the largest exact McNemar p over the seeds is 0.17. On Stanford Dogs the same pair differs by +0.05, +0.15 and −0.28 points (no claim), and Stanford's own validation picks, pairwise and random rivals, give no claim either. The verdict does not change on the provenance subsets: +0.37 points on the Stanford-origin images and +0.69 on the independent ones, both without a claim (Section 6.8).

H2, where it acts: supported on Tsinghua Dogs, refuted on Stanford Dogs. Rival-soft's second stage fixes 1.91 % of the Tsinghua test images (wrong at stage 1, right after re-ranking) and breaks 0.68 %. On the test images of the breeds in the 20 hard pairs its gain over B0 is +1.34 points, against +0.86 overall. This concentration is not specific to rival conditioning: the class query (+0.69 against +0.45), the pairwise head (+1.56 against +0.53) and the attentive probe (+0.94 against 0.00) show it too (Table 27). On Stanford Dogs fixes and breaks are equal (0.42 % each), and the gain on its own 20 pairs is −0.12 points against −0.08 overall.

H3, confusion set or contrast: supported by its criterion; the paired rule makes no claim. Random rivals reach 86.43 %, centring 86.49 % and rival 86.74 %. Random lies outside the seed spread of rival ($|86.43 - 86.74| = 0.31$ points against a larger standard deviation of 0.23) and does not exceed centring, so H3 is supported as defined in Section 5.4. The paired comparison is weaker: rival minus random is +0.31 points with one of three intervals excluding zero, so the rule itself makes no claim for rival over random; it does for rival-soft over random

(+0.69). The random control's standard deviation over five further draws of its random classes is 0.07 points on Tsinghua Dogs and 0.04 on Stanford Dogs (`top1_draw_sd`), well below the spread over seeds. On Stanford Dogs random (88.16 %) lies within the spread of rival (88.17 %), and H3 is refuted there.

H4, pairwise evidence: no difference (exploratory). Pairwise minus rival at $K = 5$ is +0.06 points on Tsinghua Dogs (per seed +0.13, −0.12, +0.15; no claim) and +0.01 on Stanford Dogs (no claim). On Stanford Dogs the two heads make almost the same decisions: the per-seed differences are +0.00, +0.01 and +0.01 points. On Tsinghua Dogs the ablation repeats the pair at other shortlist sizes: pairwise minus rival is +0.15 points at $K = 3$ (per seed +0.15, +0.13, +0.17) and +0.15 at $K = 10$ (+0.37, +0.00, +0.10). At $K = 3$ every seed is positive, but no paired interval excludes zero and the smallest McNemar p is 0.45, so the rule makes no claim at either size.

H5, cost of the shortlist: supported, with the re-ranker ahead on Tsinghua Dogs. Rival-soft beats class attention over all classes (B2) by +0.84 points, with every seed's interval above zero (+0.92, +0.65, +0.94; largest McNemar p 0.0024). On Stanford Dogs the difference is −0.09 points with no claim, which the hypothesis also counts as support (no difference). The measured cost agrees with the count of Section 4.8 (Table 25): rival-soft needs 35.3 MFLOPs per image against B2's 68.0, about half, since only the queries' share, 1/26 of B2's, shrinks with K, while both heads compute the keys and values of the 256 tokens once per image. On the T4, a batch of 256 images takes rival-soft 6.5 ms and B2 14.7 ms, with peak memory 625 against 929 MB. For a single image B2 is the faster head (0.53 against 0.86 ms); the full single-image pipeline takes 8.6 ms with rival-soft and 8.7 ms with B2, against 8.4 ms for the backbone alone.

H6, shortlist size: supported by its criterion, within the seed spread. The rival head reaches 86.85 % test top-1 at $K = 2$, 86.91 % at $K = 3$, 86.74 % at $K = 5$ and 86.52 % at $K = 10$, while its shortlist recall rises from 94.19 through 96.77 and 98.47 to 99.37 % (Table 15, Figure 4). Top-1 does not fall strictly from $K = 2$, so H6 is supported as defined in Section 5.4: accuracy is flat or peaks at a small K. The support is weak. The rise from $K = 2$ to $K = 3$ is +0.06 points and the fall from $K = 3$ to $K = 10$ is 0.39 points, against seed standard deviations of 0.05 to 0.23. The pre-registered pairs compare heads at the same K only, so the rule makes no claim across sizes. The $K = 5$ values come from the main experiment, evaluated on the Colab feature cache; the other sizes were trained and evaluated on the rebuilt cache (Section 5.3). The other two heads of the study show how fragile the support is: the class query falls strictly from $K = 2$ (86.83, 86.73, 86.71 and 86.42 %), which would refute H6 had it been stated for that head, and the pairwise head falls from $K = 3$ (87.06, 86.80 and 86.67 %).

H7, robustness: exploratory. On Tsinghua Dogs rival-soft's gain over B0 is +0.86 points on the clean view and +3.03 (blur), +0.83 (noise), +0.76 (JPEG), +1.73 (low resolution) and +2.64 points (occlusion) on the automatic corruptions; the rule gives "rival-soft > B0" on all five. The gain exceeds the clean gain under occlusion, blur and low resolution, the three views where the rival head's shortlist recall falls most (89.5, 93.9 and 96.7 %, against 98.5 % clean), and not under JPEG and noise, where it falls least (96.9 and 97.7 %). The pattern therefore does not follow the condition the prediction named, corruptions that keep the recall. On Stanford Dogs the gain lies between −0.07 and −0.23 points on every automatic view, with no claim.

6.3 Controls and the query axis

Along the query axis no step reaches a claim on Tsinghua Dogs (Table 13): centre minus class is −0.22 points, random minus centre −0.06 and rival minus random +0.31; rival-soft minus

rival is +0.38, also without a claim. Of the nine pairs of a proposed head against a control, two reach "$A > B$", both for rival-soft: over centring (+0.63) and over random rivals (+0.69, largest McNemar p 0.015). Neither rival nor pairwise beats any control. Against the baselines the rule is more favourable: each proposed head beats B2 (rival +0.46, rival-soft +0.84, pairwise +0.51) and B0 (+0.47, +0.86, +0.53), and pairwise beats B1 (+0.53), whereas the class query, a control with the same parameters, does not beat B0 (+0.45, no claim). On Stanford Dogs the only claims are rival and pairwise over the attentive probe (+1.46 and +1.47); every other pair gives no claim.

The second stage changes few decisions (Table 14). On Tsinghua Dogs the stage-1 heads of the re-rankers reach 85.89–86.19 % top-1, slightly below B0, and their shortlists hold the true breed for 98.40–98.54 % of the test images at $K = 5$. Re-ranking adds +0.37 (centred query) to +1.23 points (rival-soft). Rival-soft changes the decision for 3.56 % of the test images: it fixes 1.91 %, breaks 0.68 %, and in the remaining 0.97 % replaces one wrong breed by another. On Stanford Dogs the stage-1 heads already reach 88.19–88.36 %, re-ranking moves top-1 by −0.08 to +0.00 points, and among the changed decisions (0.49–1.10 % of the images) breaks are as frequent as fixes or slightly more frequent (fixes 0.16–0.42 %, breaks 0.21–0.42 %).

6.4 Ablations: shortlist size, training choices and token width

All three studies use Tsinghua Dogs and the learning rates of the main experiment. 22 of the ablation's 39 runs were trained on Colab and kept by the resume rule; its other 17 runs and all token-width runs were trained on Kaggle (Section 5.3). The studies use rival, not the H1-selected rival-soft, as the proposed head (Section 7).

Shortlist size. Raising K from 2 to 10 lifts the rival head's shortlist recall from 94.19 to 99.37 % but not its top-1, which stays between 86.52 and 86.91 % (Table 15, Figure 4; H6). At every size the class query and rival lie within 0.2 points of each other: rival minus class is +0.03 points at $K = 2$, +0.18 at $K = 3$ and +0.10 at $K = 10$, with no claim at any size (and +0.03 at $K = 5$). Of the three heads, pairwise is the most accurate at $K = 3$ (87.06 %), +0.15 points above rival without a claim (H4).

Training choices. No single change moves test top-1 by more than 0.15 points from the main head of its mode (Table 16). Computing the loss over the shortlist only costs rival 0.14 points; dropping the insertion of the true class into the training shortlist, or the token dropout, changes it by +0.08 and +0.02 points; replacing the soft minimum over pairs by a hard one ($t \to 0$) or a softer one ($t = 3$) changes pairwise by −0.15 and −0.03 points. These differences are smaller than the seed standard deviations of several variants (up to 0.35 points). The pre-registered pairs do not compare a variant with the main head, which belongs to another experiment, so the differences are descriptive; none of the three training choices of the main configuration is shown to matter.

Token width. Keeping 32 or 64 instead of 128 principal components of the patch tokens costs each head 0.1 to 0.4 points (Table 17): rival reaches 86.58 and 86.63 % against 86.74 %, the class query 86.49 and 86.56 against 86.71 %, and B1 86.07 and 85.90 against 86.26 %. At both narrower widths the rule finds the class query and rival ahead of B1 (+0.42 and +0.51 points at 32 dimensions, +0.66 and +0.74 at 64), which it does not find for rival at 128 dimensions (+0.47, no claim). Rival minus class is +0.08 points at both widths, with no claim.

6.5 Robustness

The two kinds of view are reported apart. Automatic views use no annotation and measure robustness of automatic inference; annotation-assisted views use the ground-truth boxes at test time and are probes only.

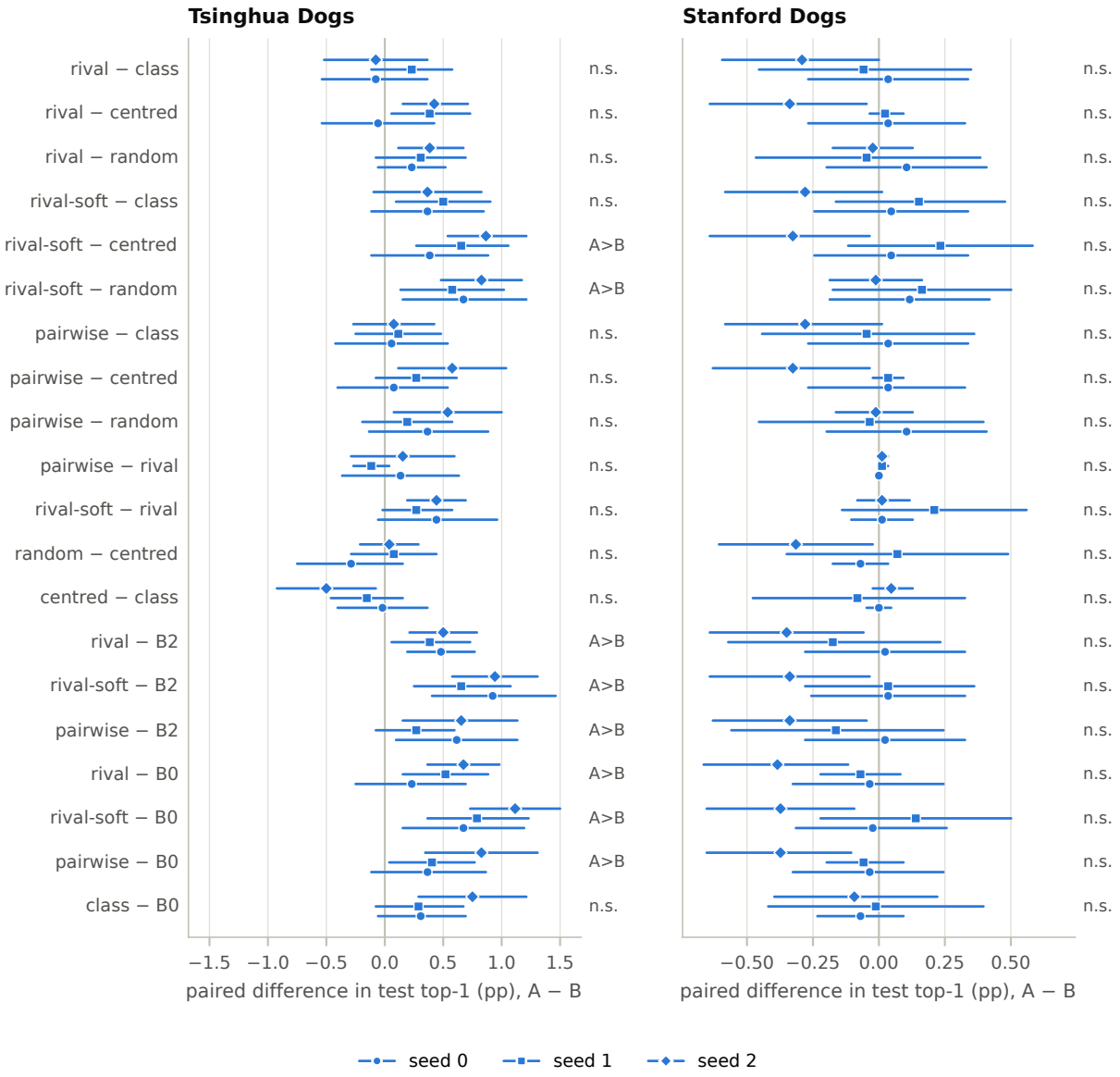


Figure 3: Forest plot of the paired comparisons on the clean test view: for each pair A − B and each seed, the difference in test top-1 (percentage points) with its paired bootstrap 95 % interval; the rule's verdict is printed on the right. A pair reaches "A > B" only when all three points lie right of zero and at least two intervals exclude zero.

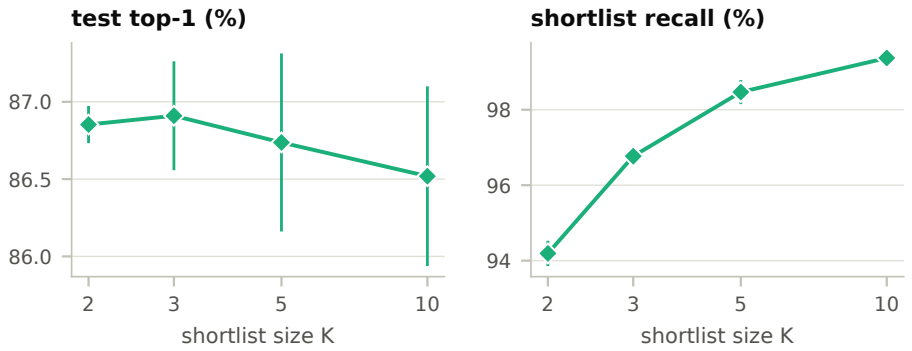


Figure 4: Test top-1 (left) and test shortlist recall (right) of the rival head against the shortlist size K on Tsinghua Dogs (logarithmic axis), mean and 95 % interval over seeds.

Table 13: Paired comparisons on the clean test view: the 22 pairs fixed in `final_all.sh`, per dataset. For each pair A − B: the mean difference in test top-1 over seeds (percentage points), the difference of each seed with its paired bootstrap 95 % interval, the largest exact McNemar p-value over the seeds, and the verdict of the rule; "no claim" means that the rule's condition holds in neither direction. The dagger marks the H1 pair.

Comparison	$\bar{\Delta}$	seed 0: Δ [95% CI]	seed 1	seed 2	max p_{McN}	Rule
Tsinghua Dogs						
rival − class	+0.03	-0.08 [-0.54, +0.37]	+0.23 [-0.12, +0.58]	-0.08 [-0.52, +0.37]	0.81	no claim
rival − centre	+0.25	-0.06 [-0.54, +0.42]	+0.38 [+0.06, +0.73]	+0.42 [+0.15, +0.71]	0.87	no claim
rival − random	+0.31	+0.23 [-0.06, +0.52]	+0.31 [-0.08, +0.69]	+0.38 [+0.12, +0.67]	0.15	no claim
rival-soft − class†	+0.41	+0.37 [-0.12, +0.85]	+0.50 [+0.10, +0.90]	+0.37 [-0.10, +0.83]	0.17	no claim
rival-soft − centre	+0.63	+0.38 [-0.12, +0.88]	+0.65 [+0.27, +1.06]	+0.87 [+0.54, +1.21]	0.16	rival-soft > centre
rival-soft − random	+0.69	+0.67 [+0.15, +1.21]	+0.58 [+0.13, +1.02]	+0.83 [+0.48, +1.17]	0.015	rival-soft > random
pairwise − class	+0.08	+0.06 [-0.42, +0.54]	+0.12 [-0.25, +0.48]	+0.08 [-0.27, +0.42]	0.88	no claim
pairwise − centre	+0.31	+0.08 [-0.40, +0.54]	+0.27 [-0.08, +0.62]	+0.58 [+0.12, +1.04]	0.81	no claim
pairwise − random	+0.37	+0.37 [-0.13, +0.88]	+0.19 [-0.19, +0.58]	+0.54 [+0.08, +1.00]	0.38	no claim
pairwise − rival	+0.06	+0.13 [-0.37, +0.63]	-0.12 [-0.27, +0.04]	+0.15 [-0.29, +0.60]	0.65	no claim
rival-soft − rival	+0.38	+0.44 [-0.06, +0.96]	+0.27 [-0.02, +0.58]	+0.44 [+0.19, +0.69]	0.1	no claim
random − centre	-0.06	-0.29 [-0.75, +0.15]	+0.08 [-0.29, +0.44]	+0.04 [-0.21, +0.29]	0.88	no claim
centre − class	-0.22	-0.02 [-0.40, +0.37]	-0.15 [-0.46, +0.15]	-0.50 [-0.92, -0.08]	1	no claim
rival − B2	+0.46	+0.48 [+0.19, +0.77]	+0.38 [+0.06, +0.73]	+0.50 [+0.21, +0.79]	0.033	rival > B2
rival-soft − B2	+0.84	+0.92 [+0.40, +1.46]	+0.65 [+0.25, +1.08]	+0.94 [+0.58, +1.31]	0.0024	rival-soft > B2
pairwise − B2	+0.51	+0.62 [+0.10, +1.13]	+0.27 [-0.08, +0.60]	+0.65 [+0.15, +1.13]	0.15	pairwise > B2
rival − B0	+0.47	+0.23 [-0.25, +0.69]	+0.52 [+0.15, +0.88]	+0.67 [+0.37, +0.98]	0.39	rival > B0
rival-soft − B0	+0.86	+0.67 [+0.15, +1.19]	+0.79 [+0.37, +1.23]	+1.12 [+0.73, +1.50]	0.014	rival-soft > B0
pairwise − B0	+0.53	+0.37 [-0.12, +0.87]	+0.40 [+0.04, +0.77]	+0.83 [+0.35, +1.31]	0.17	pairwise > B0
class − B0	+0.45	+0.31 [-0.06, +0.69]	+0.29 [-0.08, +0.67]	+0.75 [+0.29, +1.21]	0.15	no claim
rival − B1	+0.47	+0.27 [-0.17, +0.71]	+0.73 [+0.21, +1.25]	+0.42 [-0.10, +0.94]	0.27	no claim
pairwise − B1	+0.53	+0.40 [-0.15, +0.98]	+0.62 [+0.10, +1.13]	+0.58 [+0.10, +1.04]	0.18	pairwise > B1
Stanford Dogs						
rival − class	-0.10	+0.03 [-0.27, +0.34]	-0.06 [-0.45, +0.35]	-0.29 [-0.59, +0.00]	0.88	no claim
rival − centre	-0.09	+0.03 [-0.27, +0.33]	+0.02 [-0.03, +0.09]	-0.34 [-0.64, -0.05]	0.88	no claim
rival − random	+0.01	+0.10 [-0.20, +0.41]	-0.05 [-0.47, +0.38]	-0.02 [-0.17, +0.13]	0.88	no claim
rival-soft − class†	-0.03	+0.05 [-0.24, +0.34]	+0.15 [-0.16, +0.48]	-0.28 [-0.58, +0.01]	0.82	no claim
rival-soft − centre	-0.02	+0.05 [-0.24, +0.34]	+0.23 [-0.12, +0.58]	-0.33 [-0.64, -0.03]	0.82	no claim
rival-soft − random	+0.09	+0.12 [-0.19, +0.42]	+0.16 [-0.17, +0.50]	-0.01 [-0.19, +0.16]	1	no claim
pairwise − class	-0.10	+0.03 [-0.27, +0.34]	-0.05 [-0.44, +0.36]	-0.28 [-0.58, +0.01]	0.88	no claim
pairwise − centre	-0.09	+0.03 [-0.27, +0.33]	+0.03 [-0.02, +0.09]	-0.33 [-0.63, -0.03]	0.88	no claim
pairwise − random	+0.02	+0.10 [-0.20, +0.41]	-0.03 [-0.45, +0.40]	-0.01 [-0.16, +0.13]	1	no claim
pairwise − rival	+0.01	+0.00 [+0.00, +0.00]	+0.01 [+0.00, +0.03]	+0.01 [+0.00, +0.03]	1	no claim
rival-soft − rival	+0.08	+0.01 [-0.10, +0.13]	+0.21 [-0.14, +0.56]	+0.01 [-0.08, +0.12]	1	no claim
random − centre	-0.10	-0.07 [-0.17, +0.03]	+0.07 [-0.35, +0.49]	-0.31 [-0.61, -0.02]	0.79	no claim
centre − class	-0.01	+0.00 [-0.05, +0.05]	-0.08 [-0.48, +0.33]	+0.05 [-0.02, +0.13]	1	no claim
rival − B2	-0.17	+0.02 [-0.28, +0.33]	-0.17 [-0.57, +0.23]	-0.35 [-0.64, -0.06]	0.94	no claim
rival-soft − B2	-0.09	+0.03 [-0.26, +0.33]	+0.03 [-0.28, +0.36]	-0.34 [-0.64, -0.03]	0.89	no claim
pairwise − B2	-0.16	+0.02 [-0.28, +0.33]	-0.16 [-0.56, +0.24]	-0.34 [-0.63, -0.05]	0.94	no claim
rival − B0	-0.16	-0.03 [-0.33, +0.24]	-0.07 [-0.22, +0.08]	-0.38 [-0.66, -0.12]	0.87	no claim
rival-soft − B0	-0.09	-0.02 [-0.31, +0.26]	+0.14 [-0.22, +0.50]	-0.37 [-0.65, -0.09]	0.94	no claim
pairwise − B0	-0.16	-0.03 [-0.33, +0.24]	-0.06 [-0.20, +0.09]	-0.37 [-0.65, -0.10]	0.87	no claim
class − B0	-0.06	-0.07 [-0.23, +0.09]	-0.01 [-0.42, +0.40]	-0.09 [-0.40, +0.22]	1	no claim
rival − B1	+1.46	+2.10 [+1.55, +2.67]	+1.35 [+0.86, +1.84]	+0.93 [+0.52, +1.34]	1.1e-05	rival > B1
pairwise − B1	+1.47	+2.10 [+1.55, +2.67]	+1.36 [+0.87, +1.85]	+0.94 [+0.54, +1.35]	8.2e-06	pairwise > B1

Automatic views. Every head loses accuracy under every corruption on Tsinghua Dogs, most under occlusion (B0 67.7 %, rival-soft 70.3 %) and blur (74.2 and 77.3 %) and least under noise (83.9 and 84.7 %; Table 18). Rival-soft is the most accurate head on every automatic view. Of the 7 comparisons per view, the rule gives "rival-soft > B0" and "rival > B0" on all five corruptions, and "rival-soft > class" under blur, low resolution and occlusion (+1.87, +0.88, +2.05 points) but not under noise or JPEG (+0.28, +0.03). The rival head beats the class query only under blur (+0.85), as does the pairwise head (+0.78); the class query itself beats B0 under blur, noise, JPEG and low resolution but not under occlusion. The rival head's shortlist recall at $K = 5$ falls from 98.5 % (clean) to 97.7 (noise), 96.9 (JPEG), 96.7 (low resolution), 93.9 (blur) and 89.5 % (occlusion). On Stanford Dogs the heads other than B1 lie within 0.8 points of each other on every automatic view, and none of the 7 comparisons reaches a claim on any view (Table 19).

Table 14: What the second stage does, per re-ranking head and dataset: the top-1 of the head's own stage 1, the final top-1, their difference, the shortlist recall at $K = 5$, and the shares of test images whose top-1 the second stage changes, fixes (wrong to right) and breaks (right to wrong). All in % of the test images; mean and standard deviation over seeds.

Head	Stage-1 top-1	Final top-1	Gain	Recall@5	Changed	Fixed	Broken
Tsinghua Dogs							
Re-rank: class query	86.19 ± 0.18	86.71 ± 0.10	+0.52	98.52 ± 0.09	2.39 ± 0.14	1.12 ± 0.03	0.60 ± 0.10
Re-rank: centred query	86.12 ± 0.15	86.49 ± 0.17	+0.37	98.43 ± 0.04	2.35 ± 0.19	1.02 ± 0.14	0.65 ± 0.03
Re-rank: random rivals	86.05 ± 0.33	86.43 ± 0.21	+0.38	98.54 ± 0.12	1.51 ± 0.06	0.71 ± 0.06	0.33 ± 0.07
RCEA: rival	85.92 ± 0.23	86.74 ± 0.23	+0.82	98.47 ± 0.13	2.44 ± 0.29	1.28 ± 0.07	0.46 ± 0.06
RCEA: rival-soft	85.89 ± 0.20	87.12 ± 0.15	+1.23	98.44 ± 0.01	3.56 ± 0.41	1.91 ± 0.19	0.68 ± 0.21
RCEA: pairwise	86.06 ± 0.12	86.80 ± 0.12	+0.73	98.40 ± 0.01	2.62 ± 0.38	1.27 ± 0.15	0.54 ± 0.14
Stanford Dogs							
Re-rank: class query	88.36 ± 0.06	88.28 ± 0.05	-0.08	98.78 ± 0.03	0.56 ± 0.11	0.16 ± 0.02	0.24 ± 0.06
Re-rank: centred query	88.29 ± 0.05	88.26 ± 0.09	-0.02	98.81 ± 0.10	0.49 ± 0.04	0.18 ± 0.04	0.21 ± 0.03
Re-rank: random rivals	88.23 ± 0.11	88.16 ± 0.10	-0.07	98.73 ± 0.05	0.63 ± 0.10	0.20 ± 0.04	0.28 ± 0.03
RCEA: rival	88.19 ± 0.04	88.17 ± 0.12	-0.02	98.78 ± 0.12	0.76 ± 0.15	0.29 ± 0.02	0.31 ± 0.08
RCEA: rival-soft	88.25 ± 0.16	88.25 ± 0.19	+0.00	98.79 ± 0.10	1.10 ± 0.30	0.42 ± 0.09	0.42 ± 0.13
RCEA: pairwise	88.20 ± 0.03	88.18 ± 0.12	-0.02	98.78 ± 0.12	0.74 ± 0.15	0.28 ± 0.03	0.30 ± 0.08

Table 15: Shortlist size K on Tsinghua Dogs: test top-1 (%) of the class, rival and pairwise heads at $K = 2, 3, 5, 10$ ($K = 5$ from the main experiment), and the test shortlist recall of the rival head. At $K = 2$ pairwise equals rival (Fact 4) and was not trained separately.

Head	$K = 2$	$K = 3$	$K = 5$ (main)	$K = 10$
Re-rank: class query	86.83 ± 0.08	86.73 ± 0.26	86.71 ± 0.10	86.42 ± 0.34
RCEA: rival	86.85 ± 0.05	86.91 ± 0.14	86.74 ± 0.23	86.52 ± 0.23
RCEA: pairwise	= rival (Fact 4)	87.06 ± 0.15	86.80 ± 0.12	86.67 ± 0.15
Shortlist recall@K (test)	94.19	96.77	98.47	99.37

Table 16: One change at a time from the main configuration on Tsinghua Dogs: validation and test top-1 (%) of each variant and its test difference to the main head of the same mode, at the learning rate of that main head.

Variant	Val top-1	Test top-1	Δ vs main
rival: loss over the shortlist only	86.86 ± 0.12	86.60 ± 0.13	-0.14
rival: no true-class insertion in training	87.06 ± 0.05	86.81 ± 0.11	+0.08
rival: no token dropout	86.94 ± 0.05	86.76 ± 0.26	+0.02
pairwise: hard minimum over pairs ($t \to 0$)	86.99 ± 0.13	86.65 ± 0.35	-0.15
pairwise: softer minimum over pairs ($t = 3$)	86.90 ± 0.10	86.76 ± 0.09	-0.03

Table 17: Token width on Tsinghua Dogs: test top-1 (%) of B1 and of the class and rival re-rankers when the patch tokens keep 32, 64 or 128 principal components (learning rates of the main experiment).

Token PCA dims	B1	class	rival
32	86.07 ± 0.23	86.49 ± 0.17	86.58 ± 0.13
64	85.90 ± 0.33	86.56 ± 0.27	86.63 ± 0.23
128	86.26 ± 0.02	86.71 ± 0.10	86.74 ± 0.23

Annotation-assisted views (ground-truth boxes at test time; probes, not automatic inference). On Tsinghua Dogs, covering the head box costs every head most: B0 falls to 46.8 % and rival-soft to 51.1 %, and the rival head's shortlist recall to 74.1 %. Rival-soft keeps its largest lead there, +4.33 points over B0 and +3.51 over the class query, both "$A > B$". Greying the background costs B0 0.6 points and blurring it 1.7; cropping to the body box leaves B0 at 86.3 %, and cropping to the head box gives 84.4 %. Rival-soft is ahead of B0 by the rule on the grey and blurred backgrounds and the body crop (+0.79, +0.72, +0.73) and ahead of the class query on the two background views (+0.69, +0.72); none of the 7 comparisons reaches a claim on the head crop. On Stanford Dogs, which has body boxes only, no comparison reaches a claim on these views either.

Table 18: Tsinghua test top-1 (%) per view, mean over seeds. Left block: automatic views (clean and five corruptions; no annotation used). Right block: annotation-assisted views, which use the dataset's ground-truth boxes at test time and are robustness probes, not automatic inference. Last row: shortlist recall at $K = 5$ of the rival head.

	Automatic						Annotation-assisted (ground-truth boxes)				
Head	clean	blur	noise	JPEG	low-res	occlude	head occl.	bg grey	bg blur	body crop	head crop
Linear probe (B0)	86.3	74.2	83.9	81.2	79.9	67.7	46.8	85.7	84.6	86.3	84.4
Attentive probe (B1)	86.3	74.9	83.7	81.3	80.3	68.1	47.6	85.2	84.3	86.0	83.9
Class attention, all classes (B2)	86.3	74.5	83.8	81.2	80.1	67.3	46.8	85.4	84.5	86.2	84.3
Re-rank: class query	86.7	75.4	84.4	81.9	80.8	68.3	47.6	85.8	84.6	86.7	84.3
Re-rank: centred query	86.5	75.0	84.3	81.7	80.8	68.1	47.5	86.0	84.9	86.6	84.7
Re-rank: random rivals	86.4	75.4	84.0	81.5	80.5	68.4	47.7	85.8	84.7	86.4	84.4
RCEA: rival	86.7	76.2	84.3	81.9	81.1	68.5	48.7	85.9	85.1	86.7	84.5
RCEA: rival-soft	87.1	77.3	84.7	82.0	81.7	70.3	51.1	86.5	85.3	87.0	84.7
RCEA: pairwise	86.8	76.2	84.4	81.8	81.1	68.7	48.2	85.9	84.8	86.7	84.4
Shortlist recall@5 (rival)	98.5	93.9	97.7	96.9	96.7	89.5	74.1	98.4	98.2	98.5	98.0

Table 19: Stanford test top-1 (%) per view, read as Table 18; Stanford Dogs has body boxes only, so its annotation-assisted views are the grey background, the blurred background and the body crop.

	Automatic						Annotation-assisted (ground-truth boxes)		
Head	clean	blur	noise	JPEG	low-res	occlude	bg grey	bg blur	body crop
Linear probe (B0)	88.3	81.4	86.2	83.9	83.9	74.1	87.5	87.4	87.7
Attentive probe (B1)	86.7	79.9	84.3	82.1	82.3	71.6	85.6	85.5	86.2
Class attention, all classes (B2)	88.3	81.0	86.0	83.7	83.8	73.7	87.7	87.4	87.7
Re-rank: class query	88.3	81.1	86.0	83.7	83.8	73.7	87.7	87.4	87.7
Re-rank: centred query	88.3	81.3	86.1	83.7	83.8	74.2	87.6	87.3	87.7
Re-rank: random rivals	88.2	80.9	85.9	83.6	83.8	73.4	87.4	87.4	87.6
RCEA: rival	88.2	81.3	86.1	83.8	83.9	74.0	87.5	87.3	87.6
RCEA: rival-soft	88.2	81.2	86.1	83.7	83.9	73.9	87.6	87.5	87.6
RCEA: pairwise	88.2	81.2	86.1	83.8	83.9	74.0	87.5	87.3	87.6
Shortlist recall@5 (rival)	98.8	96.9	98.3	97.9	97.7	92.3	98.7	98.7	98.7

6.6 Open set

The draw holds out 30 breeds: Shiba dog, French bulldog, Pomeranian, cairn, Australian terrier, Pembroke, Bedlington terrier, Sussex spaniel, bluetick, standard schnauzer, wire-haired fox terrier, whippet, kuvasz, groenendael, malinois, Great Pyrenees, English foxhound, Eskimo dog, Gordon setter, Greater Swiss Mountain dog, soft-coated wheaten terrier, toy terrier, Italian greyhound, basset, basenji, Border collie, English springer, beagle, standard poodle and teddy. The test split then holds 4,000 images of the 100 known breeds and 1,200 unknown images (Table 20). Several held-out breeds have close relatives among the known ones: the standard schnauzer beside the giant and miniature schnauzers, the standard poodle beside the toy and miniature poodles, the Pembroke beside the Cardigan, the English springer beside the Welsh springer spaniel, and the Eskimo dog beside the Siberian husky and the malamute.

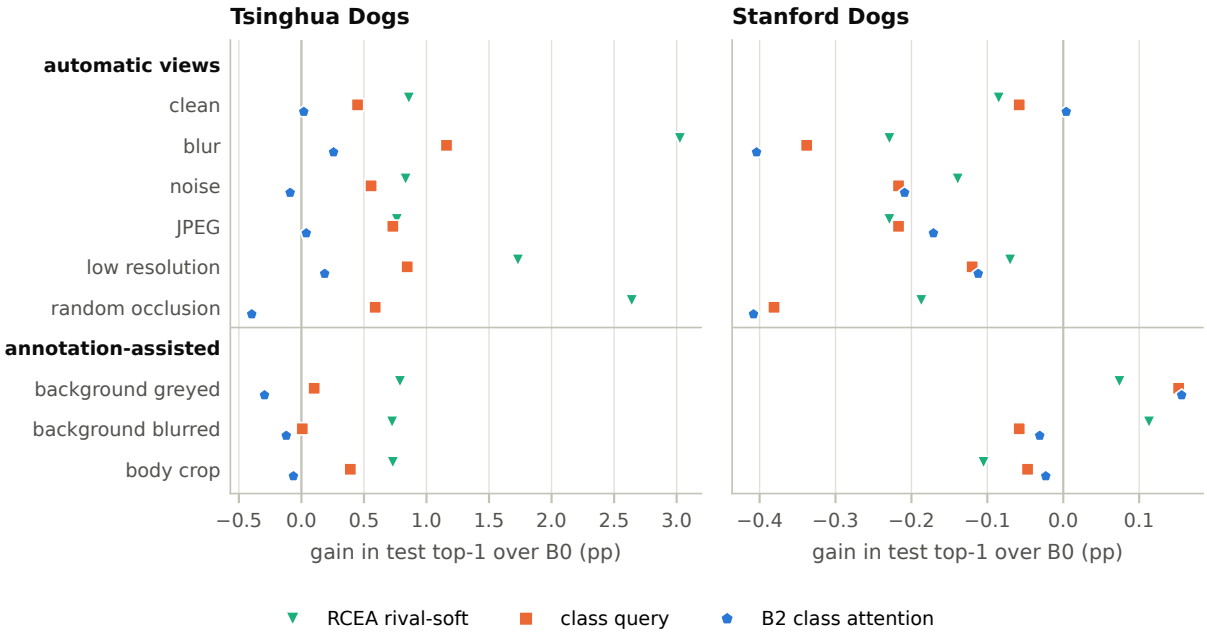


Figure 5: Gain in test top-1 over B0 (percentage points) under each view, for the H1-selected proposed head, the H1-selected control, the class query if different, and B2. Automatic views form the upper group of rows; annotation-assisted views, which use ground-truth boxes at test time, form the lower group.

The four heads are equally accurate on the known breeds (89.79–90.03 %) and separate the unknown images about equally well. With the maximum softmax probability, AUROC is 80.63–80.85 % and FPR95 65.6–67.4 %: at the threshold that keeps 95 % of the known images, about two thirds of the unknown images are accepted as well. The maximum logit is the best of the scores in the table (AUROC 81.08–81.80 %) and the energy the weakest (79.42–80.46 %); on both, the class query is ahead of the other heads by 0.7 to 1.0 points, two to three times its seed standard deviation. The second stage improves the re-rankers' energy score: their stage-1 logits give 78.54–78.89 %, below B0's 79.42 %, and their final logits 79.66–80.46 %. No paired test was pre-registered for the open set, so these differences are descriptive.

Table 20: Open set on Tsinghua Dogs: heads trained on 100 breeds; the test images of the 30 held-out breeds are the unknowns. Closed-set top-1 on the known breeds; AUROC and false-positive rate at 95 % true-positive rate for the maximum softmax probability and the energy score; AUROC for the maximum logit and for the energy of the re-rankers' stage-1 logits (%; mean and standard deviation over seeds).

Head	Top-1 (known)	MSP		Energy		MaxLogit	Energy, stage 1
		AUROC	FPR95	AUROC	FPR95	AUROC	AUROC
Linear probe (B0)	89.79 ± 0.03	80.69 ± 0.38	67.11 ± 0.60	79.42 ± 0.46	74.42 ± 0.88	81.10 ± 0.45	–
Re-rank: class query	89.83 ± 0.17	80.85 ± 0.41	65.61 ± 0.90	80.46 ± 0.37	72.53 ± 0.72	81.80 ± 0.36	78.54 ± 0.44
RCEA: rival	90.02 ± 0.18	80.65 ± 0.06	67.14 ± 0.35	79.66 ± 0.17	74.94 ± 1.02	81.08 ± 0.11	78.89 ± 0.19
RCEA: pairwise	90.03 ± 0.07	80.63 ± 0.09	67.36 ± 0.32	79.68 ± 0.24	74.69 ± 0.92	81.08 ± 0.17	78.78 ± 0.26

6.7 Transfer to Stanford Dogs

The transfer set holds 973 de-duplicated Stanford Dogs test images of 120 breeds (Table 21). Heads trained on Tsinghua Dogs reach 78.14 % (B0) to 79.75 % (rival-soft) top-1 on them and 94.38–94.96 % top-5; restricting the decision to the mapped breeds adds 0.45 to 0.76 points. The re-rankers' stage-1 heads reach 77.97–78.38 %, so re-ranking adds between +0.35 (random rivals) and +1.78 points (rival-soft). No paired test was pre-registered for the transfer set, and these differences are descriptive; the seed spread of rival-soft (0.54) is the largest of the nine heads.

Table 21: Transfer: heads trained on Tsinghua Dogs label the de-duplicated Stanford Dogs test images of the breeds with a one-to-one mapping (Section 5.10). Top-1 and top-5 of the 130-breed decision, top-1 with the decision restricted to the mapped breeds, and the stage-1 top-1 of the re-rankers. The numbers of images and breeds used are given in the text.

Head (trained on Tsinghua Dogs)	Top-1	Top-5	Top-1, mapped breeds only	Stage-1 top-1
Linear probe (B0)	78.14 ± 0.16	94.62 ± 0.21	78.83 ± 0.18	–
Attentive probe (B1)	78.38 ± 0.53	94.83 ± 0.65	79.14 ± 0.37	–
Class attention, all classes (B2)	78.31 ± 0.37	94.86 ± 0.51	78.97 ± 0.42	–
Re-rank: class query	79.00 ± 0.43	94.52 ± 0.12	79.72 ± 0.43	78.38 ± 0.30
Re-rank: centred query	78.59 ± 0.12	94.38 ± 0.30	79.34 ± 0.18	78.08 ± 0.21
Re-rank: random rivals	78.49 ± 0.21	94.76 ± 0.37	79.07 ± 0.16	78.14 ± 0.43
RCEA: rival	78.86 ± 0.36	94.93 ± 0.26	79.38 ± 0.46	78.21 ± 0.31
RCEA: rival-soft	79.75 ± 0.54	94.96 ± 0.21	80.20 ± 0.66	77.97 ± 0.52
RCEA: pairwise	79.17 ± 0.16	94.76 ± 0.27	79.79 ± 0.16	78.28 ± 0.16

6.8 Provenance subsets

Every head is 5.7 to 6.5 points less accurate on the 1,362 independent test images than on the 3,115 Stanford-origin ones (B0: 81.38 against 87.70 %; rival-soft: 82.70 against 88.44 %; Table 22). The gap is consistent with the Stanford-origin images being ILSVRC-2012 training images (Section 3.5), but the subsets also differ in their breeds and photographs, so it is not a measure of leakage alone. Rival-soft is the most accurate head on both subsets. The H1 pair gives no claim on either (+0.37 and +0.69 points), so the H1 verdict holds on the independent images. Rival-soft's lead over B0 holds on both subsets (+0.74 and +1.32, "$A > B$" on each); its lead over B2 holds on the independent subset (+1.39, "$A > B$") but not on the Stanford-origin one (+0.70, no claim). The rival head beats B2 on the independent subset (+0.78) but not B0 there (+0.71, no claim). Of the nine pairs of a proposed head and a control, one reaches a claim on a subset: rival-soft over random rivals on the independent images (+1.03). Rival-soft over centring, which reaches one on the full split, does not on either subset (+0.66 and +0.83).

6.9 Other backbones

ResNet-50 finished on Colab before the GPU quota ended; CLIP ViT-B/16 and ConvNeXt-T were finished on Kaggle (Section 5.3). Every head of the three chose the learning rate $3 \cdot 10^{-4}$, the lowest of the grid, as every head did in both DINOv2 experiments (Table 24).

With ResNet-50 features the four heads are 0.7 to 1.0 points below their DINOv2 counterparts: B0 85.59 %, class query 85.76 %, rival 85.85 % and pairwise 85.79 % (Table 23). Rival minus class is +0.09 points (per seed +0.04, +0.08, +0.15), with no claim, and so are rival minus B0 (+0.26), class minus B0 (+0.17), pairwise minus class (+0.03) and pairwise minus rival (−0.06).

With CLIP ViT-B/16 features every head is 7 to 8 points below DINOv2: B0 78.24 %, class query 79.51 %, rival 79.83 % and pairwise 79.62 %. Re-ranking gains more here than on any other backbone, and the rule finds both the class query (+1.27 points) and rival (+1.59) ahead of B0. Rival minus class is +0.32 points (per seed −0.08, +0.63, +0.40), with no claim, and so are pairwise minus class (+0.11) and pairwise minus rival (−0.21).

ConvNeXt-T gives the most accurate heads of the study: B0 86.95 %, class query 87.65 %, rival 87.86 % and pairwise 87.89 %, 0.7 to 1.1 points above their DINOv2 counterparts. The rule again finds the class query (+0.71 points) and rival (+0.91) ahead of B0 and makes no claim between the re-rankers: rival minus class is +0.21 points (per seed +0.21, +0.02, +0.38), pairwise minus class +0.23 and pairwise minus rival +0.03. This ranking of backbones has to be read with the audit of Section 3.5: ConvNeXt-T was trained with labels on ILSVRC-2012, which contains every Stanford Dogs image and therefore 3,115 of the 5,200 Tsinghua test images, whereas DINOv2 saw these images without labels. The provenance subsets were evaluated for

Table 22: Tsinghua test top-1 (%) on the provenance subsets (Section 3.4): all 5,200 images, the 3,115 images with a confirmed Stanford Dogs copy (all of them ILSVRC-2012 training images), and the 1,362 images with no candidate match. Lower block: paired differences (percentage points, mean over seeds) with the rule's outcome on each subset; the H1 pair first (its "§8" is the design document's Section 8, i.e. Eq. (12)). The independent subset is not a random sample: compare heads within a column, not columns.

Head	All test images (5,200)	Stanford-origin (3,115)	Independent (1,362)
Linear probe (B0)	86.26 ± 0.21	87.70 ± 0.14	81.38 ± 0.68
Attentive probe (B1)	86.26 ± 0.02	87.61 ± 0.12	81.60 ± 0.31
Class attention, all classes (B2)	86.28 ± 0.29	87.75 ± 0.18	81.30 ± 0.36
Re-rank: class query	86.71 ± 0.10	88.07 ± 0.07	82.01 ± 0.19
Re-rank: centred query	86.49 ± 0.17	87.78 ± 0.07	81.86 ± 0.45
Re-rank: random rivals	86.43 ± 0.21	87.81 ± 0.23	81.67 ± 0.28
RCEA: rival	86.74 ± 0.23	88.10 ± 0.18	82.08 ± 0.38
RCEA: rival-soft	87.12 ± 0.15	88.44 ± 0.29	82.70 ± 0.35
RCEA: pairwise	86.80 ± 0.12	88.05 ± 0.15	82.21 ± 0.28
Paired differences in pp, mean over seeds (rule of §8)			
rival-soft − class†	+0.41 (n.s.)	+0.37 (n.s.)	+0.69 (n.s.)
rival-soft − B2	+0.84 (>)	+0.70 (n.s.)	+1.39 (>)
rival-soft − B0	+0.86 (>)	+0.74 (>)	+1.32 (>)
rival − class	+0.03 (n.s.)	+0.03 (n.s.)	+0.07 (n.s.)
rival − centre	+0.25 (n.s.)	+0.32 (n.s.)	+0.22 (n.s.)
rival − random	+0.31 (n.s.)	+0.29 (n.s.)	+0.42 (n.s.)
pairwise − class	+0.08 (n.s.)	-0.02 (n.s.)	+0.20 (n.s.)
pairwise − rival	+0.06 (n.s.)	-0.05 (n.s.)	+0.12 (n.s.)
rival − B2	+0.46 (>)	+0.35 (n.s.)	+0.78 (>)
rival − B0	+0.47 (>)	+0.40 (n.s.)	+0.71 (n.s.)
class − B0	+0.45 (n.s.)	+0.36 (n.s.)	+0.64 (n.s.)

the main experiment only, so how much of ConvNeXt-T's lead comes from this overlap is not known; ResNet-50, trained with labels on the same images, is below DINOv2.

Across the four backbones rival minus class lies between +0.03 and +0.32 points and never reaches a claim. Rival beats B0 by the rule on three backbones (all but ResNet-50), the class query on two (CLIP and ConvNeXt-T).

Table 23: Frozen backbones on Tsinghua Dogs: test top-1 (%) of B0 and of the class, rival and pairwise re-rankers, the mean rival-minus-class difference over seeds, and the rule's outcome for that pair. Each backbone has its own learning-rate selection. The supervised ImageNet backbones were trained with labels on every Stanford Dogs image (Section 3.5).

Backbone	Pre-training	B0	class	rival	pairwise	$\bar{\Delta}$ (rival−class)	Rule
DINOv2 ViT-B/14 (reg.)	self-sup., LVD-142M	86.26 ± 0.21	86.71 ± 0.10	86.74 ± 0.23	**86.80 ± 0.12**	+0.03	no claim
CLIP ViT-B/16	image-text, WIT-400M	78.24 ± 0.10	79.51 ± 0.14	**79.83 ± 0.49**	79.62 ± 0.50	+0.32	no claim
ConvNeXt-T	supervised IN-1k	86.95 ± 0.12	87.65 ± 0.23	87.86 ± 0.06	**87.89 ± 0.07**	+0.21	no claim
ResNet-50	supervised IN-1k	85.59 ± 0.11	85.76 ± 0.01	**85.85 ± 0.05**	85.79 ± 0.03	+0.09	no claim

6.10 Efficiency

The watcher measured the cost from 03:53:03 to 03:53:43 UTC on a Tesla T4 of the Kaggle session, after the last training and evaluation job had ended (Table 25; torch 2.11.0, median of 50 timed passes after 10 warm-up passes). The measured FLOPs are twice the multiply-accumulates counted in Table 6: 35.3 M per image for B1 and for the class, centred, random, rival and rival-soft heads, 39.2 M for pairwise, 68.0 M for B2 and 0.4 M for B0. The frozen backbone needs 47.1 GFLOPs and 8.4 ms per image at batch 1, so a re-ranker adds under 0.1 % to its FLOPs, and B2 0.14 %. At batch 1 the heads alone take 0.09 ms (B0) to 1.03 ms (pairwise),

Table 24: Learning rate chosen for each head on validation top-1 (seed 0) from $\{3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}\}$, for each experiment that runs its own selection.

Head	tsinghua main	stanford main	tsinghua resnet50	tsinghua convnext tiny	tsinghua clip vitb16
Linear probe (B0)	0.0003	0.0003	0.0003	0.0003	0.0003
Attentive probe (B1)	0.0003	0.0003	–	–	–
Class attention, all classes (B2)	0.0003	0.0003	–	–	–
Re-rank: class query	0.0003	0.0003	0.0003	0.0003	0.0003
Re-rank: centred query	0.0003	0.0003	–	–	–
Re-rank: random rivals	0.0003	0.0003	–	–	–
RCEA: rival	0.0003	0.0003	0.0003	0.0003	0.0003
RCEA: rival-soft	0.0003	0.0003	–	–	–
RCEA: pairwise	0.0003	0.0003	0.0003	0.0003	0.0003

and the full single-image pipeline (backbone, token PCA, head) 7.8 to 9.1 ms for every head. These differences between heads are of the size of the timing noise: the pipeline with B0 (7.8 ms) measured faster than the backbone alone. At batch 256, B2 takes 14.7 ms, pairwise 8.1 ms and the other re-rankers 6.4 to 6.6 ms, so a re-ranker needs 43 to 55 % of B2's time; peak memory is 929 MB for B2 and 620 to 666 MB for the re-rankers.

Table 25: Cost on a Tesla T4 (Kaggle): parameters, FLOPs per image (a multiply-add counts as 2 FLOPs), median latency of one pass at batch 1 and of one pass at batch 256 (the backbone at batch 64), peak GPU memory at that batch size, and the latency of the full single-image pipeline (backbone, token PCA, head). The backbone row is the frozen ViT-B/14 in fp16.

Part	Params	FLOPs / image	ms (b=1)	ms (b=256)	Peak MB (b=256)	Pipeline ms (b=1)
Backbone (frozen, fp16)	85,727,232	47.1 G	8.4	243 (b=64)	765	–
Linear probe (B0)	202,882	0.4 M	0.09	0.1	410	7.8
Attentive probe (B1)	404,866	35.3 M	0.43	6.4	644	8.3
Class attention, all classes (B2)	302,211	68.0 M	0.53	14.7	929	8.7
Re-rank: class query	302,211	35.3 M	0.62	6.4	620	8.4
Re-rank: centred query	302,211	35.3 M	0.70	6.4	621	8.6
Re-rank: random rivals	302,211	35.3 M	0.75	6.5	623	8.5
RCEA: rival	302,211	35.3 M	0.76	6.6	624	8.2
RCEA: rival-soft	302,211	35.3 M	0.86	6.5	625	8.6
RCEA: pairwise	302,211	39.2 M	1.03	8.1	666	9.1

6.11 Where the attention falls

The measurement uses seed 0 on the Tsinghua test split (Table 26). The body boxes cover 79.9 % of the image area on average and the head boxes 29.4 %. The class and centred queries concentrate their attention: 51.7 and 47.5 % of the mass falls in the head box (lift 2.54 and 2.63), and their maps have the lowest relative entropy (0.785 and 0.727). The three contrastive queries spread their attention almost evenly: relative entropy 0.962–0.976, body lift 0.94–1.02 and head lift 0.92–1.24 (rival-soft: 80.2 % of the mass in the body box, 32.1 % in the head box). The attentive probe lies between them (head lift 1.72), and class attention over all classes puts less mass on the head than its share of the area (lift 0.80). The heads that re-rank best are thus not the ones whose attention is most concentrated on the dog. With an almost uniform map, the pooled evidence is close to the mean token, so the score is close to the mean per-patch margin, the lower end of the range in Fact 3. Figure 6 shows six images (the rival and class heads, seed 0). The class query's maps are more peaked than the rival head's: their bright spots fall on the dog in the poodle and Norfolk terrier rows and on patches of the floor in the Pomeranian and Shiba rows, while the rival head's maps vary less across the image. Within each head, the maps of the first and the second candidate look alike.

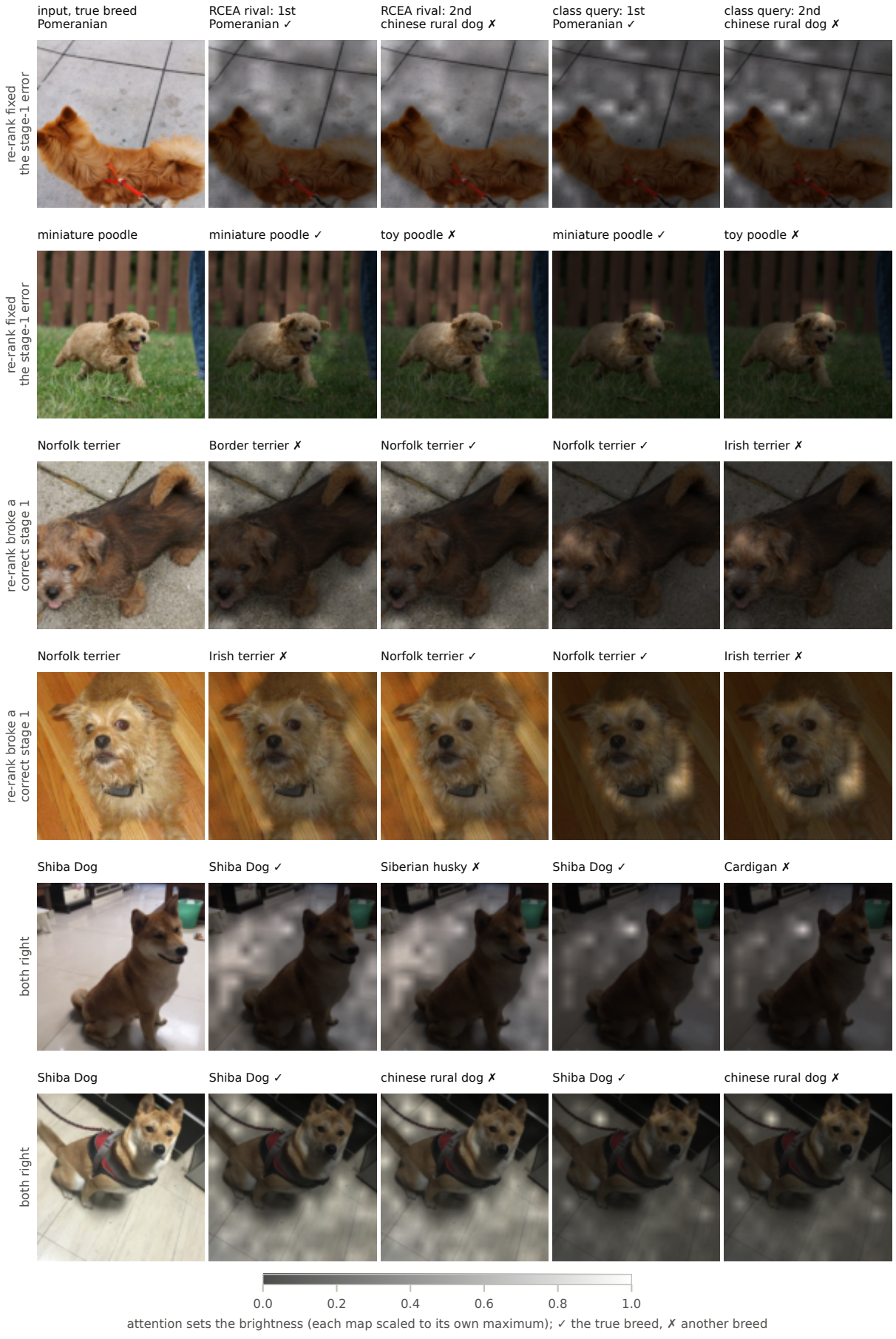


Figure 6: Attention of the two highest-scoring final candidates of the rival and class heads (seed 0) on Tsinghua test images: two images of each kind (the rival head fixed a stage-1 error, broke a correct stage-1 answer, or both were right), the first by image id, not hand-picked. Brightness follows the attention, each map scaled to its own maximum; a tick marks the true breed and a cross another breed.

Table 26: Attention against the boxes (Tsinghua test split, seed 0): top-1 of each head, the attention mass inside the ground-truth body and head boxes, the boxes' share of the image area, the lift (mass divided by area; 1 means no preference) and the entropy of the map relative to a uniform map. The boxes serve this measurement only; no head uses them.

		Body box			Head box			
Head	Top-1	mass	area	lift	mass	area	lift	H/H_{unif}
Attentive probe (B1)	86.2	86.3	79.9	1.15	38.9	29.4	1.72	0.972
Class attention, all classes (B2)	86.0	74.1	79.9	0.90	24.8	29.4	0.80	0.977
Re-rank: class query	86.6	90.2	79.9	1.23	51.7	29.4	2.54	0.785
Re-rank: centred query	86.6	88.7	79.9	1.22	47.5	29.4	2.63	0.727
RCEA: rival	86.5	76.2	79.9	0.95	26.8	29.4	0.93	0.969
RCEA: rival-soft	87.0	80.2	79.9	1.02	32.1	29.4	1.24	0.962
RCEA: pairwise	86.7	76.2	79.9	0.94	26.8	29.4	0.92	0.976

6.12 Per-breed changes and hard pairs

Against B0, rival-soft raises the test accuracy of 70 of the 130 Tsinghua breeds, lowers it for 29 and leaves 31 unchanged (mean over seeds; Figure 7). The largest gains are for the miniature poodle (+8.3 points), the Japanese spitz and the Pembroke (+6.7 each), and the silky terrier and the malamute (+5.8 each); the largest losses for the Yorkshire terrier (−5.0), the Siberian husky and the Australian terrier (−3.3 each). On Stanford Dogs 46 breeds gain, 47 lose and 27 are unchanged.

The 20 hard pairs of Tsinghua Dogs (most confused by B0 on validation, seed 0) are led by toy poodle and teddy, with 217 confusions, followed by toy poodle and Bichon Frise (35), Chinese rural dog and papillon (24), Bichon Frise and teddy (21), and Pomeranian and Chinese rural dog (20); eight of the 20 pairs involve the Chinese rural dog. On the test images of these breeds rival-soft makes 152.7 within-pair errors against 160.3 for B0, and the pairwise head the fewest, 149.0 (Table 27). Stanford's 20 pairs hold 2 to 5 validation confusions each; on them no head gains more than +0.04 points over B0 (Table 28).

Table 27: The 20 breed pairs most confused by B0 (seed 0) on validation, Tsinghua Dogs: test errors within these pairs, accuracy on the test images of the breeds in the pairs, overall top-1, and the gains over B0 on the pair breeds and overall (percentage points).

Head	Pair errors	Acc. on pair breeds	Top-1	Gain on pairs	Gain overall
Linear probe (B0)	160.3	71.70	86.26	+0.00	+0.00
Attentive probe (B1)	153.0	72.65	86.26	+0.94	+0.00
Class attention, all classes (B2)	159.7	71.99	86.28	+0.29	+0.02
Re-rank: class query	156.3	72.39	86.71	+0.69	+0.45
Re-rank: centred query	155.0	72.03	86.49	+0.33	+0.22
Re-rank: random rivals	156.7	72.10	86.43	+0.40	+0.17
RCEA: rival	152.3	72.86	86.74	+1.16	+0.47
RCEA: rival-soft	152.7	73.04	87.12	+1.34	+0.86
RCEA: pairwise	149.0	73.26	86.80	+1.56	+0.53

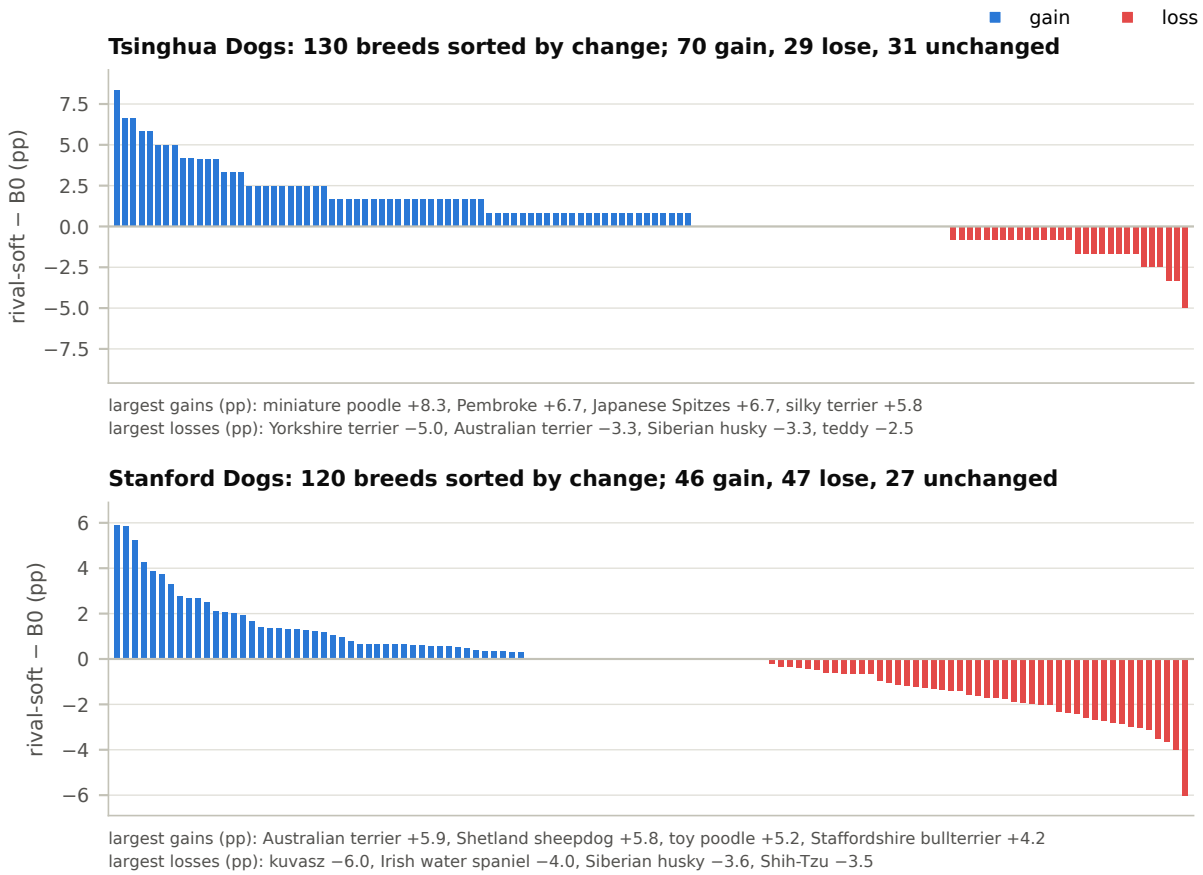


Figure 7: Per-breed change in test accuracy, the H1-selected proposed head minus B0 (mean over seeds), with the breeds sorted by the change; the largest gains and losses are named under each panel.

Table 28: The same as Table 27 for Stanford Dogs, with its own 20 pairs.

Head	Pair errors	Acc. on pair breeds	Top-1	Gain on pairs	Gain overall
Linear probe (B0)	216.7	81.30	88.33	+0.00	+0.00
Attentive probe (B1)	244.0	78.67	86.71	-2.63	-1.62
Class attention, all classes (B2)	225.0	81.34	88.34	+0.04	+0.00
Re-rank: class query	227.0	81.27	88.28	-0.03	-0.06
Re-rank: centred query	216.3	81.15	88.26	-0.14	-0.07
Re-rank: random rivals	229.3	81.03	88.16	-0.27	-0.17
RCEA: rival	218.7	81.11	88.17	-0.19	-0.16
RCEA: rival-soft	223.3	81.18	88.25	-0.12	-0.08
RCEA: pairwise	218.7	81.11	88.18	-0.19	-0.15

7 Limitations and threats to validity

Frozen features only. Every head is trained on cached features of a frozen backbone. Whether an effect of the query survives end-to-end fine-tuning, where the backbone could learn the same distinctions itself, is not tested: end-to-end fine-tuning with the RCEA head was not run in this study, and the question stays open.

Pre-training overlap. Every Stanford Dogs image, and through its copy every Stanford-origin Tsinghua test image, is an ILSVRC-2012 training image; the supervised ImageNet backbones saw them with labels, and DINOv2 saw the ImageNet images through ImageNet-22k without labels (an inference from its published data composition, not a count). Absolute accuracies are therefore optimistic for every backbone. Comparisons between heads share the features, but a

head could still draw more on memorised images than another. The independent subset is the cleaner check, but it is not a random sample. Checks recommended in our literature dossier were not run: a backbone pre-trained without dog classes [12, 86], and test sets outside the pre-training data, such as the dog classes of ImageNetV2 [89], the dog breeds of Oxford-IIIT Pet [88] after removing known near-duplicates [87], or Dogs-in-the-Wild [21], which was built without ImageNet images.

Selection at the edge of the grid. In both main experiments and the ResNet-50 study every head chose the lowest learning rate of the grid. The best rate may lie below it. The grid was fixed before training and was not widened afterwards, since widening it after seeing where the choices fell would be a second selection round; all heads are affected in the same way, but not necessarily to the same degree.

Three seeds. Three seeds limit the power of any test across seeds. The decision rule therefore leans on paired per-image tests within each seed and requires agreement in sign across all three; a real but small difference can end as "no claim".

Token compression. The token heads read 128 of 256 principal components fitted on 4,000 training images; the token-width runs vary this only downwards (32 and 64). Raw tokens and wider projections were not tried.

Annotation-assisted views. Five views use the dataset's boxes at test time. They answer what happens when the background or the head is removed or isolated, not how the system performs as an automatic recogniser, and they are reported apart. The Tsinghua boxes are semi-automatic (detector proposals with human correction of low-confidence cases [10]); Stanford Dogs has body boxes only.

Open set. The held-out breeds are unknown to the heads, not to the backbone, and the token PCA was fitted without labels on training images that include them. One draw of 30 held-out breeds is used.

Which proposed head the secondary studies use. The secondary studies (shortlist size, training-choice ablations, token widths, open set, other backbones) and the attention figure were configured before the H1 pick, with rival as the representative proposed head, and were not changed when rule 1 picked rival-soft. Their tables name the head they use; they do not test the H1-selected head.

Baselines that were not run. The comparison set is internal: equal-parameter controls and three frozen-feature baselines. A SAC-style joint re-classifier [4], a training-free top-K log-softmax-gradient re-scorer (Fact 2), a head–body two-crop baseline [35], CSQA-Net [36] and fine-tuned FGVC methods were not re-implemented. The literature numbers of Table 1 come from other backbones, resolutions and protocols and are not comparable.

Splits and labels. The Tsinghua test split is its official validation split, which is the benchmark's de facto test set; our validation split is carved from the official training split. Training is long-tailed while the test split is balanced. Near-duplicate pairs with conflicting labels (113 and 48) were not relabelled. The hard pairs come from one run (B0, seed 0) on validation data.

Compute and procedure. The main experiments ran on one free-tier Colab T4 with 2 virtual CPUs; code reached the virtual machine as hashed bundles typed into notebook cells. Google Drive was mounted only from about 21:13 UTC, when the owner had approved it; from then on `/content/work` (runs, checkpoints, logs) was mirrored to Drive every 10 minutes, but the feature caches stayed on the virtual machine's disk and were lost when the usage limit ended the runtime at about 22:14 UTC, which also cut off the secondary experiments. These were finished on Kaggle, with another torch and cuDNN version and on rebuilt feature caches. The resume rule bounds the effect on the runs kept from Colab (at most 3 of 6,515 validation predictions changed), but the $K = 5$ column of the shortlist-size study comes from the Colab cache and the other sizes from the Kaggle one. The Stanford Dogs archives came from Internet Archive snapshots, accepted on the official checksums. While the launch of the Tsinghua locked evaluation was being checked (21:05 UTC), its first log lines, the clean-view test top-1 of a few seed runs, were visible on screen. Three changes to the analysis code followed between 21:10 and 21:20 UTC: the comparison table lists all 22 pairs that `final_all.sh` computes instead of 14, and rival-soft was added to the per-breed data and to the attention statistics. The log records them as following from the validation pick alone; no comparison pair, decision rule or selection was changed. The secondary experiments were frozen and evaluated by a watcher, and their freeze summaries entered the log after their evaluations (Section 5.3). The study has one implementation and no external replication.

8 Conclusion

We described rival-conditioned evidence attention, a re-ranking head whose queries are contrasted against the image's own shortlist, and four facts that place it among class-query, re-ranking and contrastive-explanation methods. Equal-parameter controls isolate the query, a leakage audit fixes provenance subsets of the Tsinghua test split, and a protocol frozen before any test evaluation states how each hypothesis is judged. Judged by that protocol, the central hypothesis is refuted. On Tsinghua Dogs, contrasting a candidate's query against its own shortlist does not beat the same head with a plain class query: rival-soft leads by 0.41 points in the mean, positive in all three seeds but with only one seed's paired interval excluding zero. The verdict is the same on the test images that match no Stanford Dogs image. What the rule does support is that shortlist re-ranking on frozen DINOv2 features helps on Tsinghua Dogs: rival-soft fixes about 2.8 test images for each one it breaks, gains most on the most confused breed pairs, and beats both a linear probe and class attention over all 130 classes, also on the independent images. The class query, a control with exactly the same parameters, is only 0.41 points behind it and not separable from it by the rule, so the rule does not attribute these gains to the rival conditioning itself. On Stanford Dogs, whose images are all ILSVRC-2012 training images, no head improves on the linear probe. In the one seed measured, the attention of the contrastive heads is close to uniform, which suggests that what they add acts through the score rather than through where they look; this was not tested. The secondary studies, finished under the same protocol, agree with this reading. Between $K = 2$ and $K = 10$ the rival head's accuracy stays within 0.4 points while its shortlist recall rises from 94 to 99 %, which meets the criterion of H6 only within the seed spread. No training choice moves it by more than 0.15 points, and narrower tokens cost 0.1 to 0.4 points. On four frozen backbones the rule never separates rival from the class query, while re-ranking beats the linear probe on three of them. Every head detects held-out breeds about equally well, and per image a re-ranker costs about half the FLOPs of class attention over all classes. ConvNeXt-T features give the highest accuracy, but its labelled pre-training overlaps the test images, so the comparison of backbones is not clean.

A Environment

Compute. One Google Colab runtime with an NVIDIA T4 (15 GB), 12 GB of RAM, 2 virtual CPUs and about 113 GB of disk, on the free tier; no paid resource was used. Code reached the virtual machine as an xz-compressed, base64-encoded tar bundle typed into notebook cells in pieces of at most 9 KB; each piece was checked by SHA-256 before it ran and the whole payload by MD5 on the machine (`dist/make_bundle.py`, `dist/split_cell.py`). Text files were normalised to LF line endings in the bundle.

Compute for the secondary studies. One free-tier Kaggle notebook session with two NVIDIA T4 GPUs (15,360 MiB each, driver 580.178.04) and 31 GB of RAM, from 02:04 to 03:54 UTC on 9 November (1 h 50 min of the account's 30 weekly GPU hours); no paid resource was used. The repository and the resume archive of the work folder reached it as files of a private Kaggle dataset uploaded by the owner, and the notebook checked both against their SHA-256 before use. Software: Python 3.13.16; torch 2.11.0+cu130 (CUDA 13.0, cuDNN 92700); timm 1.0.31; numpy 2.1.3; scipy 1.16.3; scikit-learn 1.6.1; pandas 2.3.3. Its full `pip freeze` (938 lines) is `requirements-kaggle-full.txt`. The session packed its results into two archives and wrote their SHA-256: `kaggle_results_small.tar` (6,113,280 bytes, `bc84641d...`; every result file except checkpoints and prediction arrays) and `kaggle_results_full.tar` (1,205,340,160 bytes, `82fbbd4f...`). The digest, tables and figures were made from the small archive, which matched its hash on arrival. The notebook later ran a second time, from 05:43 to 07:31 UTC (about 1 h 48 min more of the weekly hours), from the same inputs. It repeated every step: the drift checks, the training, the freezes and the test evaluations, and made the same selections. Every validation and test metric of every run came out the same; only timings, time stamps and the drift check's random draws for the random-rival control differ. Its 93 frozen checkpoints have the SHA-256 that the first session's freezes recorded. No reported number comes from this second run. The Drive mirror of the work folder holds the first session's result files, merged from its small archive, and the 315 checkpoints and prediction arrays of the second run's full archive, each checked before it was copied: a checkpoint against the SHA-256 recorded at its freeze, and a prediction array by recomputing from it the validation or test metrics the first session recorded. The 16 learning-rate grid checkpoints at the rates not chosen have no recorded hash and were checked through their validation predictions (`scripts/merge_rerun_files.py`; record in `results/kaggle_merge`).

Software on the runtime. As pinned in `requirements-colab.txt` from `pip freeze` on the virtual machine: Python 3.13.16; torch 2.14.1+cu130 (CUDA 13.0, cuDNN 92800); torchvision 0.29.1+cu130; timm 1.0.31; numpy 2.4.6; scipy 1.16.3; scikit-learn 1.6.1; pandas 2.2.3; matplotlib 3.10.0; pillow 11.3.0; PyYAML 6.0.3; GPU driver 580.82.07. The full `pip freeze` of the runtime, 723 lines (722 requirements and pip's comment on the editable install of this repository), is `requirements-colab-full.txt`. It was written to the Drive mirror at 21:28:56 UTC and recovered from there on 9 November through Drive's text view; the rebuilt file has the Drive copy's size, 14,256 bytes (SHA-256 `e9b05fe8...`).

Local machine. Unit tests and smoke runs only: Python 3.12.15 in a virtual environment, torch 2.15.0+cpu, torchvision 0.29.2+cpu, timm 1.0.31, numpy 2.5.4, pillow 13.0.0, pandas 3.1.1, scikit-learn 1.9.1, scipy 1.18.1, matplotlib 3.11.2, pytest 9.2.0, PyYAML 6.0.3, with 2 CPU threads (`research_log/env_setup.log`). The repository's 42 tests pass locally and on a Colab virtual machine; the 36 that existed before the Kaggle session also passed on Kaggle.

B Data sources and hashes

Table 29 lists the archives as recorded in the research log and the download logs. Tsinghua Dogs publishes no checksums.

Table 29: Data archives and auxiliary files. Stanford Dogs hashes are SHA-256 values equal to those TensorFlow Datasets records for the official release (abbreviated as in the log).

File	Bytes	Source	Hash
Stanford `images.tar`	793,579,520	Internet Archive snapshot 20250724000413 of the official URL	`a443fc2f...ec58`
Stanford `annotation.tar`	21,852,160	snapshot 20250724000413 of the official URL	`662ed36f...0fd`
Stanford `lists.tar`	481,280	snapshot 20241221112710 of the official URL	`34b47cac...d6c`
Tsinghua `TrainValSplit.zip`	198,475	cg.cs.tsinghua.edu.cn/ThuDogs	not published
Tsinghua `low-annotations.zip`	37,326,330	cg.cs.tsinghua.edu.cn/ThuDogs	not published
Tsinghua `low-resolution.zip`	2,664,983,287	cloud.tsinghua.edu.cn (official link); finished 16:52:55 UTC	not published
ILSVRC-2012 `train.txt`		BVLC Caffe `caffe_ilsvrc12.tar.gz` (17,858,008 bytes); 1,281,167 lines	`55a0a92c27626550...`
evaluation code bundle		14 files, typed into notebook cells (19:45–20:11 UTC)	MD5 `9c663e63...`

The three Tsinghua Dogs archives, as kept on the Drive mirror for rebuilding the caches, have the SHA-256 `9dd6543c...` (`low-resolution.zip`), `89d1c44d...` (`low-annotations.zip`) and `2281c9fc...` (`TrainValSplit.zip`). The repository keeps the full values, and `scripts/kaggle_run.sh` refuses to run unless a new download matches them; the file is `configs/expected_sha256_tsinghua.txt`. The Kaggle session's downloads (02:05–02:09 UTC on 9 November) matched all three. The 15 files of the result digest are listed with their SHA-256 and sizes in `results/digest/MANIFEST.txt`; the copy in the repository matches all 15.

C Command sequence

The study runs from three Colab notebooks and one Kaggle notebook; these are the commands they run, from the repository root (paths under `/content`; on Kaggle, `kaggle_run.sh` links the same paths). Steps 4 and 5 are repeated for every experiment: `tsinghua_main`, `stanford_main`, `tsinghua_ablate`, `tsinghua_openset`, `tsinghua_tok32`, `tsinghua_tok64`, `tsinghua_resnet50`, `tsinghua_convnext_tiny`, `tsinghua_clip_vitb16`.

```
# 1. data, manifests, within-dataset audit (download_data.sh, prepare_data.py)
bash scripts/colab_data.sh all
# 2. feature caches (Stanford: own PCA and the Tsinghua PCA basis); cross-dataset audit
bash scripts/extract_all.sh tsinghua dinov2_vitb14_reg \
    /content/data/raw/tsinghua_dogs/low-resolution.zip 256 224
bash scripts/pipeline_stanford.sh
#    Stanford Dogs against the ILSVRC-2012 training list, by file id
mkdir -p /content/aux && cd /content/aux
wget http://dl.caffe.berkeleyvision.org/caffe_ilsvrc12.tar.gz
tar -xzf caffe_ilsvrc12.tar.gz train.txt synsets.txt && cd -
python -I scripts/ilsvrc_overlap.py /content/aux /content/work
# 3. training: learning-rate grid on seed 0, then seeds 0-2; then the job queues
bash scripts/run_experiment.sh configs/experiments/tsinghua_main.yaml
bash scripts/queue.sh configs/queues/a.txt   # ablation, open-set split and runs, token widths
bash scripts/queue.sh configs/queues/b.txt   # stanford_main, other backbones' feature caches
```

```
bash scripts/queue.sh configs/queues/c.txt   # other backbones' heads (from queue A, 21:24 UTC)
# 4. freeze model selection (per experiment), then record it in research_log/research_log.md
python -I scripts/freeze.py --experiment tsinghua_main
# 5. locked test evaluation (per experiment), after the freeze is recorded
bash scripts/final_all.sh tsinghua_main > /content/work/logs/final_tsinghua_main.log 2>&1
#    secondary experiments: the watcher runs steps 4-5 unattended, then steps 6-9
setsid nohup bash scripts/auto_final.sh > /content/work/logs/auto_final.log 2>&1 < /dev/null &
# 6. efficiency, with nothing else on the GPU (reads no test data)
python -I scripts/efficiency_report.py --config configs/experiments/tsinghua_main.yaml
# 7. digest: every result into results/digest/ with a SHA-256 manifest
python -I scripts/digest.py --curves tsinghua_main/linear_s0,tsinghua_main/rr_class_s0,\
tsinghua_main/rr_rival_s0,tsinghua_main/rr_rival_soft_s0,tsinghua_main/rr_pairwise_s0
cd results/digest && awk '{print $1 "  " $3}' MANIFEST.txt | sha256sum -c - && cd -
# 8. tables and numbers.json
python -I scripts/make_tables.py --digest results/digest --out paper/generated
# 9. figures
python -I scripts/make_figures.py --gen paper/generated --out paper/figures \
    --explain results/tsinghua_main/explain/shown_maps.json \
    --images data/raw/tsinghua_dogs/low-resolution.zip
#    after the Colab GPU quota: the six secondary experiments on Kaggle (9 November)
bash scripts/make_resume_bundle.sh /content/drive/MyDrive/DogBreed-PhD-Research \
    /content/resume_out                     # on Colab (CPU): resume.tar, expected_sha256.txt
mkdir -p /kaggle/working/DogBreed-PhD-Research    # Kaggle notebook, dataset attached
tar -xzf $(find /kaggle/input -name repo_tgz.bin | head -1) \
    -C /kaggle/working/DogBreed-PhD-Research
bash /kaggle/working/DogBreed-PhD-Research/scripts/kaggle_run.sh  # data check, drift
                                            # check, queues, watcher (steps 3-6)
bash scripts/ingest_kaggle.sh /content/kaggle_in  # back on Colab: hashes, merge, 7-9
python -I scripts/merge_rerun_files.py --archive kaggle_results_full.tar \
    --work /content/work --mirror /content/drive/MyDrive/DogBreed-PhD-Research/work \
    --record /content/drive/MyDrive/DogBreed-PhD-Research/kaggle_runs/20261109_095050
                                            # 9 November: checkpoints and predictions
                                            # of the second run, checked file by file
# 10. paper
apt-get install -y latexmk texlive-latex-recommended texlive-latex-extra \
    texlive-fonts-recommended texlive-pictures
cd paper && latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The ablation, open-set and token-width configurations borrow the learning rates of `tsinghua_main`, so queue A waits until that experiment has finished. `final_all.sh` fixes the views, the comparison pairs and their order per experiment (Section 5.5); every step that reads a test split refuses to run without a matching freeze.

D Hyperparameters

Table 30: Hyperparameters, from the experiment configurations (`configs/experiments/*.yaml`) and the training defaults of the code (`train_heads.py`, `heads.py`, `rival_attention.py`).

Setting	Value
Backbone (main)	DINOv2 ViT-B/14 with 4 registers, LVD-142M, frozen; 224 px
Evaluation transform	shorter side to 256 (bicubic), centre crop 224, ImageNet normalisation (CLIP: its own)
Tokens	$N = 256$, registers dropped; PCA with 256 components fitted on 4,000 training images (seed 0); first 128 used (32, 64 in the token-width runs); stored as float16
Global descriptor	$[\mathrm{CLS}; \bar{x}]$, $D_g = 1536$
Cached views	clean (training, validation, test); flipped (training); the test views of Table 9
Seeds	0, 1, 2
Learning rate	grid $\{3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}\}$ on seed 0 (main experiments, other backbones); borrowed from `tsinghua_main` (ablation, open set, token width)
Optimiser	AdamW; weight decay 0.05 on parameters with at least two dimensions, 0 otherwise
Schedule	linear warm-up over 1 epoch, then cosine decay to 0 at epoch 30
Epochs, batch, patience	at most 30; 256; 8 (early stopping on validation top-1, ties by validation NLL; best epoch kept)
Loss	label smoothing 0.1; $\alpha = 1$ for two-stage heads (B2 and re-rankers)
Regularisation	token dropout 0.1 (keys masked, at least one kept); dropout 0; gradient clipping at norm 1; random choice of the clean or flipped view per training image
Precision	heads in fp32 on the T4 (bf16 only on GPUs of compute capability 8 or more); backbone in fp16 autocast
Re-ranking head	$K = 5$; $d = 256$; $H = 4$; $d_h = 64$; $\tau = 1$; $T = 1$; $t = 1$; true-class insertion on; attention bias on; γ initialised to 1; $e_j \sim \mathcal{N}(0, 0.02^2)$; $W \sim \mathcal{N}(0, 0.01^2)$, $b^0 = 0$
B1	$M = 4$ queries, $d = 256$, $H = 4$
Ablation values	$K \in \{2, 3, 10\}$ (rival, class; pairwise at 3 and 10); $t \in \{0, 3\}$ (pairwise); loss over the shortlist only; true-class insertion off; token dropout 0
Open set	30 of 130 breeds held out (seed 0); heads B0, class, rival, pairwise
GPU memory	training tokens kept on the GPU up to 85 % of its memory (7.7 GB for Tsinghua Dogs)

E Freeze mechanism and timeline

`scripts/freeze.py` refuses a head with fewer than the expected number of finished seeds and refuses a second freeze of the same experiment unless forced. It writes `experiments/<exp>/freeze.json` with the time, each head's rate and, per seed run, the validation metrics, best epoch and SHA-256 of `best.pt`. `final_eval.py`, `ood_eval.py`, `transfer_eval.py`, `subset_eval.py` and `explain.py` check the freeze and the hashes before they read a test split. For the two main experiments the printed summary was entered in the research log before their test evaluation; for the secondary experiments the watcher keeps it in `logs/freeze_<exp>.log` and it is copied into the log afterwards (Section 5.3). The log's own rule is that nothing below its freeze heading is edited after the first test evaluation. Table 31 gives the recorded events.

The freeze summaries of the six Kaggle experiments were copied into the research log from `logs/freeze_<exp>.log` after their evaluations, as for `tsinghua_resnet50`.

Table 31: Timeline of 8 and 9 November 2026 (UTC), from the research log and, for the Kaggle session, from its logs (`kaggle_run.log`, `auto_final_kaggle.log`, `drift_*.log`).

Time	Event
16:52:55	Tsinghua Dogs image archive downloaded (resumed transfers)
17:46 / 18:08	manifests and within-dataset audits of Tsinghua Dogs / Stanford Dogs
18:19:40	training of `tsinghua_main` started (learning-rate grid, then three seeds)
18:21:59	Tsinghua feature extraction finished
18:22:10	Stanford pipeline started (caches, transfer cache, cross-dataset audit)
18:47:39	cross-dataset audit stopped on a descriptor lookup for a split-excluded image; fixed and tested
19:03:19	evaluation scripts and job queues A and B deployed
19:10:16–19:12:50	cross-dataset audit re-run on the CPU
19:45–20:11	evaluation code locked: fixed draws for the random control at evaluation; freeze enforced in code
20:13:39	`stanford_main` frozen (27 runs, 9 heads)
20:18	analysis rules 1–7 recorded (Appendix F); no test prediction existed
20:18:56–20:50:58	locked test evaluation of `stanford_main`
20:59:37 / 21:03:42	training of `tsinghua_main` finished / frozen (27 runs, 9 heads); H1 pair recorded
21:05:02–21:28:25	locked test evaluation of `tsinghua_main`; its first log lines were visible on screen at 21:05 while its launch was checked
21:10–21:20	analysis code made to follow the validation pick: all 22 pairs in the comparison table, rival-soft added to the per-breed data and the attention statistics; no pair, rule or selection changed
about 21:13	Google Drive mounted on the GPU runtime; `/content/work` mirrored to it every 10 minutes
21:24:27	queue C (other backbones) started beside queue A
21:28	Stanford Dogs counted against the ILSVRC-2012 training list
21:42	watcher declared that freezes and evaluates each secondary experiment when its training ends (`auto_final.sh`)
21:56:00	`tsinghua_resnet50` frozen by the watcher (12 runs, 4 heads)
21:56:02–21:57:39	locked test evaluation of `tsinghua_resnet50`
22:09:58	last push of `/content/work` to Drive
about 22:14	GPU runtime ended by the Colab usage limit; `tsinghua_ablate` (22 of 39 runs trained) and `tsinghua_convnext_tiny` (6 runs) cut off, neither frozen and no test split of either read
23:35	move to Kaggle declared in the research log, with the resume rule (Section 5.3)
9 November, Kaggle session with two T4 GPUs	
02:04:41	repository and resume archive checked by SHA-256 in the notebook; `kaggle_run.sh` started
02:05:14–02:09:22	Tsinghua archives downloaded, all three matching the Colab hashes; queues started on both GPUs
02:23:18–02:23:49	drift check of `tsinghua_main` (24 heads) and `tsinghua_ablate` (22 runs): at most 0.0003; ablation resumed with its finished runs
02:32:03–02:32:06	drift check of `tsinghua_convnext_tiny` (6 runs): 0; resumed with its finished runs
02:40:25	`tsinghua_convnext_tiny` frozen (12 seed runs, 4 heads); evaluated 02:40:26–02:40:59
03:06:03	`tsinghua_ablate` frozen (39 runs, 13 heads); evaluated 03:06:04–03:07:13
03:08:16	`tsinghua_clip_vitb16` frozen (12, 4); evaluated 03:08:16–03:08:48
03:23:51	`tsinghua_openset` frozen (12, 4); evaluated 03:23:52–03:24:10
03:38:12	`tsinghua_tok32` frozen (9, 3); evaluated 03:38:13–03:38:40
03:52:42	`tsinghua_tok64` frozen (9, 3); evaluated 03:52:43–03:53:03
03:53:03–03:53:43	efficiency report, after every training and evaluation job had ended
03:54:23	results packed (`kaggle_results_small.tar`, `kaggle_results_full.tar`) with their SHA-256
05:43:20–07:30:59	on Kaggle: a second run of the notebook from the same inputs; the six experiments frozen again (06:19:00–07:29:24) and test-evaluated, with the same selections and metrics
05:53:31–05:54:06	on Colab: small archive checked and merged into the runtime's work folder only (`LOCAL_ONLY=1`); digest, tables and figures made (`scripts/ingest_kaggle.sh`)
09:08	second run's full archive uploaded to Drive; its SHA-256 is not the recorded one
09:50:50–09:56:24	on Colab: small archive merged into the Drive mirror, old folders kept; digest, tables and figures identical (`scripts/ingest_kaggle.sh`); the second run's 315 checkpoints and prediction arrays, each checked against the first session's records, copied to the work folder and the mirror (`scripts/merge_rerun_files.py`)

F Pre-registered analysis rules

Recorded at 20:18 UTC, before any test prediction existed, and implemented in `scripts/make_tables.py` (`best_by_val` and `hypotheses`):

1. *H1.* The best proposed mode and the best control are the tags with the highest mean validation top-1 over their three seed runs on Tsinghua Dogs; ties go to the lower mean validation NLL. Stanford Dogs repeats the same pair ("same sign on Stanford"); its own validation picks are reported beside it and not used for the verdict. H2, H5 and H7 use the same proposed head.
2. *H3.* "Within the seed spread" means that $|\text{random} - \text{rival}|$ is at most the larger of the two test top-1 standard deviations over seeds.
3. *The random control* is evaluated with fixed draws (seeded with the run's seed); `top1_draw_sd` is reported.
4. *Hard pairs (H2)* are the 20 breed pairs most confused by B0, seed 0, in its validation predictions.
5. *Provenance subsets* are fixed by the overlap audit: Stanford-origin 3,115 images, independent 1,362.
6. *Comparisons* are exactly the pairs in `scripts/final_all.sh`; each is judged per seed (paired bootstrap with 10,000 resamples; exact McNemar) and by Eq. (12).
7. Every experiment's configuration, learning-rate grid and seeds were fixed before its training. Nothing is re-selected after a test evaluation.

G Statistical procedures

Intervals over seeds. For $n = 3$ seed values with mean m and standard deviation s ($n - 1$ in the denominator), the 95 % interval is $m \pm t_{0.975,n-1}\, s/\sqrt{n}$ with $t_{0.975,2} = 4.303$.

Paired bootstrap. Per seed, the test images are resampled with replacement 10,000 times (generator seed 0) and the difference of the two heads' top-1 is computed on each resample; the interval is formed by the 2.5th and 97.5th percentiles.

Exact McNemar test. With n_{10} images right under A and wrong under B and n_{01} the reverse, the two-sided p-value is that of a binomial test of n_{10} successes in $n_{10} + n_{01}$ trials with probability 1/2.

Temperature scaling. One temperature per run, fitted on its validation logits by minimising the NLL over $\log T$ with L-BFGS (learning rate 0.1, at most 200 iterations, strong Wolfe line search).

Calibration and selective prediction. ECE with 15 equal-width confidence bins; Brier score; the area under the risk–coverage curve of the maximum softmax probability and the risk at 90 % coverage.

Open set. AUROC with the known images as positives; FPR95 is the share of unknown images scoring at or above the 5 % quantile of the known images' scores.

Hard pairs. From B0, seed 0, on validation: for every unordered breed pair $\{a, b\}$ the number of images of a predicted as b plus the reverse; the 20 largest. On the test split, the pair errors and the accuracy on test images whose true breed occurs in one of the pairs.

H Training curves

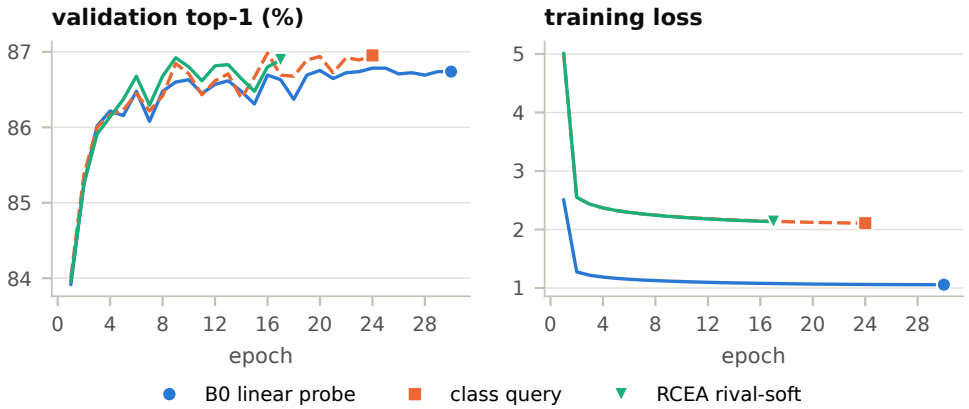


Figure 8: Validation top-1 per epoch (left) and training loss (right) for seed 0 of B0, the class query and the H1-selected proposed head on Tsinghua Dogs; the marker shows the last epoch run (early stopping).

I What is verified and what is still open

Table 32: Status of the material this draft depends on, on 9 November 2026 at 10:15 UTC.

Item	Status
Data sizes, within-dataset audit, cross-dataset overlap, ILSVRC-2012 count	recorded in the research log; used in Section 3
Freezes of `stanford_main` and `tsinghua_main`, analysis rules	recorded before the respective test evaluations
Parameter counts and MACs	computed from the code (Table 6)
Test results of all nine experiments	in the repository (`results/digest`, 15 files, each matching its SHA-256 in `MANIFEST.txt`); used in Section 6
Freeze of `tsinghua_resnet50`	made by the watcher at 21:56:00 UTC, before its test evaluation; summary copied into the research log afterwards
Resume rule and drift check	rule recorded at 23:35 UTC on 8 November; checks on Kaggle at most 0.0003 (`results/drift`); finished runs of `tsinghua_ablate` and `tsinghua_convnext_tiny` kept
Freezes of `tsinghua_ablate`, `tsinghua_convnext_tiny`, `tsinghua_openset`, `tsinghua_tok32`, `tsinghua_tok64`, `tsinghua_clip_vitb16`	made by the watcher on Kaggle, each before its test evaluation (Table 31); summaries copied into the research log afterwards
Efficiency measurements	made on a Kaggle T4 after the last experiment (`results/digest/efficiency.csv`)
Result archives of the Kaggle session	small archive (all result files) checked by SHA-256 and merged, into the Drive mirror too; the checkpoints and prediction arrays on the mirror come from the second run's full archive (`44670d50...`), each checked against the first session's records before it was copied (`results/kaggle_merge`); the first session's full archive was not needed
Second Kaggle run	05:43–07:31 UTC, from the same inputs; every selection and every validation and test metric the same as the first session's; no reported number comes from it
Tsinghua archive checksums, digest manifest, full `pip freeze` of both runtimes	in the repository (`configs/expected_sha256_tsinghua.txt`, `results/digest/MANIFEST.txt`, `requirements-colab-full.txt`, `requirements-kaggle-full.txt`)
Fine-tuning, dog-free pre-training, ImageNet-free test sets, SAC-style and training-free baselines	not run

References

[1] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=a68SUt6zFt.

[2] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024.

[3] Ke Zhu and Jianxin Wu. Residual attention: A simple but effective method for multi-label recognition. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 184–193, 2021. doi: 10.1109/ICCV48922.2021.00025.

[4] Tuong Do, Huy Tran, Erman Tjiputra, Quang D. Tran, and Anh Nguyen. Fine-grained visual classification using self assessment classifier. In *2024 IEEE Conference on Artificial Intelligence (CAI)*, pages 597–602, 2024. doi: 10.1109/CAI59869.2024.00117.

[5] Chunshui Cao, Xianming Liu, Yi Yang, Yinan Yu, Jiang Wang, Zilei Wang, Yongzhen Huang, Liang Wang, Chang Huang, Wei Xu, Deva Ramanan, and Thomas S. Huang. Look and think twice: Capturing top-down visual attention with feedback convolutional neural networks. In *2015 IEEE International Conference on Computer Vision (ICCV)*, pages 2956–2964, 2015. doi: 10.1109/ICCV.2015.338.

[6] Giang Nguyen, Valerie Chen, Mohammad Reza Taesiri, and Anh Totti Nguyen. PCNN: Probable-class nearest-neighbor explanations improve fine-grained image classification accuracy for AIs and humans. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=OcFjqiJ98b.

[7] Reza Esfandiarpoor and Stephen H. Bach. Follow-up differential descriptions: Language models resolve ambiguities for image classification. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=g6rZtxaXRm.

[8] Jindong Gu, Yinchong Yang, and Volker Tresp. Understanding individual decisions of CNNs via contrastive backpropagation. In *Computer Vision – ACCV 2018*, Lecture Notes in Computer Science, pages 119–134, 2019. doi: 10.1007/978-3-030-20893-6_8.

[9] Brian Kenji Iwana, Ryohei Kuroki, and Seiichi Uchida. Explaining convolutional neural networks using softmax gradient layer-wise relevance propagation. In *2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*, pages 4176–4185, 2019. doi: 10.1109/ICCVW.2019.00513.

[10] Ding-Nan Zou, Song-Hai Zhang, Tai-Jiang Mu, and Min Zhang. A new dataset of dog breed images and a benchmark for fine-grained classification. *Computational Visual Media*, 6(4):477–487, 2020. doi: 10.1007/s41095-020-0184-6.

[11] Aditya Khosla, Nityananda Jayadevaprakash, Bangpeng Yao, and Li Fei-Fei. Novel dataset for fine-grained image categorization: Stanford dogs. In *First Workshop on Fine-Grained Visual Categorization (FGVC), IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Colorado Springs, CO, June 2011.

[12] Jonathan Krause, Benjamin Sapp, Andrew Howard, Howard Zhou, Alexander Toshev, Tom Duerig, James Philbin, and Li Fei-Fei. The unreasonable effectiveness of noisy data for fine-grained recognition. In *Computer Vision – ECCV 2016*, Lecture Notes in Computer Science, pages 301–320, 2016. doi: 10.1007/978-3-319-46487-9_19.

[13] Tsung-Yu Lin, Aruni RoyChowdhury, and Subhransu Maji. Bilinear CNN models for fine-grained visual recognition. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1449–1457, 2015. doi: 10.1109/ICCV.2015.170.

[14] Ze Yang, Tiange Luo, Dong Wang, Zhiqiang Hu, Jun Gao, and Liwei Wang. Learning to navigate for fine-grained classification. In *Computer Vision – ECCV 2018*, Lecture Notes in Computer Science, pages 438–454. Springer, 2018. doi: 10.1007/978-3-030-01264-9_26.

[15] Tao Hu, Honggang Qi, Qingming Huang, and Yan Lu. See better before looking closer: Weakly supervised data augmentation network for fine-grained visual classification, 2019. arXiv preprint arXiv:1901.09891.

[16] Robert van der Klis, Stephan Alaniz, Massimiliano Mancini, Cassio F. Dantas, Dino Ienco, Zeynep Akata, and Diego Marcos. PDiscoNet: Semantically consistent part discovery for fine-grained recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1866–1876, 2023. doi: 10.1109/ICCV51070.2023.00179.

[17] Ruoyi Du, Dongliang Chang, Ayan Kumar Bhunia, Jiyang Xie, Zhanyu Ma, Yi-Zhe Song, and Jun Guo. Fine-grained visual classification via progressive multi-granularity training of jigsaw patches. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, pages 153–168. Springer, 2020. doi: 10.1007/978-3-030-58565-5_10.

[18] Yue Chen, Yalong Bai, Wei Zhang, and Tao Mei. Destruction and construction learning for fine-grained image recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5152–5161, 2019. doi: 10.1109/CVPR.2019.00530.

[19] Dongliang Chang, Yifeng Ding, Jiyang Xie, Ayan Kumar Bhunia, Xiaoxu Li, Zhanyu Ma, Ming Wu, Jun Guo, and Yi-Zhe Song. The devil is in the channels: Mutual-channel loss for fine-grained image classification. *IEEE Transactions on Image Processing*, 29:4683–4695, 2020. doi: 10.1109/TIP.2020.2973812.

[20] Wei Luo, Xitong Yang, Xianjie Mo, Yuheng Lu, Larry Davis, Jun Li, Jian Yang, and Ser-Nam Lim. Cross-X learning for fine-grained visual categorization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8241–8250, 2019. doi: 10.1109/ICCV.2019.00833.

[21] Ming Sun, Yuchen Yuan, Feng Zhou, and Errui Ding. Multi-attention multi-class constraint for fine-grained image recognition. In *Computer Vision – ECCV 2018*, Lecture Notes in Computer Science, pages 834–850, 2018. doi: 10.1007/978-3-030-01270-0_49.

[22] Yongming Rao, Guangyi Chen, Jiwen Lu, and Jie Zhou. Counterfactual attention learning for fine-grained visual categorization and re-identification. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1005–1014, 2021. doi: 10.1109/ICCV48922.2021.00106.

[23] Ju He, Jie-Neng Chen, Shuai Liu, Adam Kortylewski, Cheng Yang, Yutong Bai, and Changhu Wang. TransFG: A transformer architecture for fine-grained recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 852–860, 2022. doi: 10.1609/aaai.v36i1.19967.

[24] Jun Wang, Xiaohan Yu, and Yongsheng Gao. Feature fusion vision transformer for fine-grained visual categorization. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2021.

[25] Yunqing Hu, Xuan Jin, Yin Zhang, Haiwen Hong, Jingfeng Zhang, Yuan He, and Hui Xue. RAMS-Trans: Recurrent attention multi-scale transformer for fine-grained image recognition. In *Proceedings of the 29th ACM International Conference on Multimedia*, pages 4239–4248, 2021. doi: 10.1145/3474085.3475561.

[26] Hongbo Sun, Xiangteng He, and Yuxin Peng. SIM-Trans: Structure information modeling transformer for fine-grained visual categorization. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 5853–5861, 2022. doi: 10.1145/3503161.3548308.

[27] Qin Xu, Jiahui Wang, Bo Jiang, and Bin Luo. Fine-grained visual classification via internal ensemble learning transformer. *IEEE Transactions on Multimedia*, 25:9015–9028, 2023. doi: 10.1109/TMM.2023.3244340.

[28] Haowei Zhu, Wenjing Ke, Dong Li, Ji Liu, Lu Tian, and Yi Shan. Dual cross-attention learning for fine-grained visual categorization and object re-identification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4682–4692, 2022. doi: 10.1109/CVPR52688.2022.00465.

[29] Po-Yung Chou, Yu-Yung Kao, and Cheng-Hung Lin. Fine-grained visual classification with high-temperature refinement and background suppression, 2023. arXiv preprint arXiv:2303.06442.

[30] Qishuai Diao, Yi Jiang, Bin Wen, Jia Sun, and Zehuan Yuan. MetaFormer: A unified meta framework for fine-grained recognition, 2022. arXiv preprint arXiv:2203.02751.

[31] Xin Jiang, Hao Tang, Junyao Gao, Xiaoyu Du, Shengfeng He, and Zechao Li. Delving into multimodal prompting for fine-grained visual classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 2570–2578, 2024. doi: 10.1609/aaai.v38i3.28034.

[32] Yongji Zhang, Siqi Li, Kuiyang Huang, Yue Gao, and Yu Jiang. H^3Former: Hypergraph-based semantic-aware aggregation via hyperbolic hierarchical contrastive loss for fine-grained visual classification, 2025.

[33] Sangwon Kim, Jaeyeal Nam, and Byoung Chul Ko. ViT-NeT: Interpretable vision transformers with neural tree decoder. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 11162–11172, 2022. URL https://proceedings.mlr.press/v162/kim22g.html.

[34] Jiongxin Liu, Angjoo Kanazawa, David Jacobs, and Peter Belhumeur. Dog breed classification using part localization. In *Computer Vision – ECCV 2012*, Lecture Notes in Computer Science, pages 172–185, 2012. doi: 10.1007/978-3-642-33718-5_13.

[35] Noman Khan, Afnan, Mi Young Lee, and Jakyoung Min. Head-body guided deep learning framework for dog breed recognition. *Computers, Materials & Continua*, 85(2):2935–2958, 2025. doi: 10.32604/cmc.2025.069058.

[36] Qin Xu, Sitong Li, Jiahui Wang, Bo Jiang, Bin Luo, and Jinhui Tang. Context-semantic quality awareness network for fine-grained visual categorization. *Pattern Recognition*, 170: 112033, 2026. doi: 10.1016/j.patcog.2025.112033.

[37] Duy M. Le, Bao Q. Bui, Anh Tran, Cong Tran, and Cuong Pham. Enhancing fine-grained image classification through attentive batch training. *ICT Express*, 12(2):319–323, 2026. doi: 10.1016/j.icte.2025.10.002.

[38] Shilong Liu, Lei Zhang, Xiao Yang, Hang Su, and Jun Zhu. Query2Label: A simple transformer way to multi-label classification, 2021.

[39] Tal Ridnik, Gilad Sharir, Avi Ben-Cohen, Emanuel Ben-Baruch, and Asaf Noy. ML-Decoder: Scalable and versatile classification head. In *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 32–41, 2023. doi: 10.1109/WACV56688.2023.00012.

[40] Lian Xu, Wanli Ouyang, Mohammed Bennamoun, Farid Boussaid, and Dan Xu. Multi-class token transformer for weakly supervised semantic segmentation. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4300–4309, 2022. doi: 10.1109/CVPR52688.2022.00427.

[41] Ronghui You, Zihan Zhang, Ziye Wang, Suyang Dai, Hiroshi Mamitsuka, and Shanfeng Zhu. AttentionXML: Label tree-based attention-aware deep model for high-performance extreme multi-label text classification. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pages 5812–5822, 2019.

[42] Dipanjyoti Paul, Arpita Chowdhury, Xinqi Xiong, Feng-Ju Chang, David Carlyn, Samuel Stevens, Kaiya L. Provost, Anuj Karpatne, Bryan Carstens, Daniel Rubenstein, Charles Stewart, Tanya Berger-Wolf, Yu Su, and Wei-Lun Chao. A simple interpretable transformer for fine-grained image classification and analysis. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=bkdWThqE6q.

[43] Xiaokang Chen, Mingyu Ding, Xiaodi Wang, Ying Xin, Shentong Mo, Yunhao Wang, Shumin Han, Ping Luo, Gang Zeng, and Jingdong Wang. Context autoencoder for self-supervised representation learning. *International Journal of Computer Vision*, 132(1): 208–223, 2024. doi: 10.1007/s11263-023-01852-4.

[44] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=QaCCuDfBk2.

[45] Alaaeldin El-Nouby, Michal Klein, Shuangfei Zhai, Miguel Angel Bautista, Alexander Toshev, Vaishaal Shankar, Joshua M. Susskind, and Armand Joulin. Scalable pre-training of large autoregressive image models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 12371–12384, 2024. URL https://proceedings.mlr.press/v235/el-nouby24a.html.

[46] Bill Psomas, Dionysis Christopoulos, Eirini Baltzi, Ioannis Kakogeorgiou, Tilemachos Aravanis, Nikos Komodakis, Konstantinos Karantzalos, Yannis Avrithis, and Giorgos Tolias. Attention, please! revisiting attentive probing through the lens of efficiency. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://openreview.net/forum?id=PXo0gtT7Al.

[47] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015.

[48] Karim Ahmed, Mohammad Haris Baig, and Lorenzo Torresani. Network of experts for large-scale image categorization. In *Computer Vision – ECCV 2016*, Lecture Notes in Computer Science, pages 516–532, 2016. doi: 10.1007/978-3-319-46478-7_32.

[49] Zhicheng Yan, Hao Zhang, Robinson Piramuthu, Vignesh Jagadeesh, Dennis DeCoste, Wei Di, and Yizhou Yu. HD-CNN: Hierarchical deep convolutional neural networks for large scale visual recognition. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2740–2748, 2015. doi: 10.1109/ICCV.2015.314.

[50] Shaokang Yang, Shuai Liu, Cheng Yang, and Changhu Wang. Re-rank coarse classification with local region enhanced features for fine-grained image recognition, 2021.

[51] Canshi Wei. Enhancing fine-grained image classifications via cascaded vision language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1857–1871, 2024. doi: 10.18653/v1/2024.findings-emnlp.102.

[52] Ziyu Liu, Zeyi Sun, Yuhang Zang, Wei Li, Pan Zhang, Xiaoyi Dong, Yuanjun Xiong, Dahua Lin, and Jiaqi Wang. RAR: Retrieving and ranking augmented MLLMs for visual recognition. *IEEE Transactions on Image Processing*, 35:388–401, 2026. doi: 10.1109/TIP.2025.3644175.

[53] Yulin Wang, Kangchen Lv, Rui Huang, Shiji Song, Le Yang, and Gao Huang. Glance and focus: a dynamic approach to reducing spatial redundancy in image classification. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[54] Yulin Wang, Zhaoxi Chen, Haojun Jiang, Shiji Song, Yizeng Han, and Gao Huang. Adaptive focus for efficient video recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 16229–16238, 2021. doi: 10.1109/ICCV48922.2021.01594.

[55] Yulin Wang, Yang Yue, Yuanze Lin, Haojun Jiang, Zihang Lai, Victor Kulikov, Nikita Orlov, Humphrey Shi, and Gao Huang. AdaFocus V2: End-to-end training of spatial dynamic networks for video recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20030–20040, 2022. doi: 10.1109/CVPR52688.2022.01943.

[56] Yulin Wang, Yang Yue, Xinhong Xu, Ali Hassani, Victor Kulikov, Nikita Orlov, Shiji Song, Humphrey Shi, and Gao Huang. AdaFocusV3: On unified spatial-temporal dynamic video recognition. In *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science, pages 226–243. Springer, 2022. doi: 10.1007/978-3-031-19772-7_14.

[57] Jianming Zhang, Zhe Lin, Jonathan Brandt, Xiaohui Shen, and Stan Sclaroff. Top-down neural attention by excitation backprop. In *Computer Vision – ECCV 2016*, Lecture Notes in Computer Science, pages 543–559, 2016. doi: 10.1007/978-3-319-46493-0_33.

[58] Jianming Zhang, Sarah Adel Bargal, Zhe Lin, Jonathan Brandt, Xiaohui Shen, and Stan Sclaroff. Top-down neural attention by excitation backprop. *International Journal of Computer Vision*, 126(10):1084–1102, 2018. doi: 10.1007/s11263-017-1059-x.

[59] Mohit Prabhushankar, Gukyeong Kwon, Dogancan Temel, and Ghassan AlRegib. Contrastive explanations in neural networks. In *2020 IEEE International Conference on Image Processing (ICIP)*, pages 3289–3293, 2020. doi: 10.1109/ICIP40778.2020.9190927.

[60] Pei Wang and Nuno Vasconcelos. SCOUT: Self-aware discriminant counterfactual explanations. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8978–8987, 2020. doi: 10.1109/CVPR42600.2020.00900.

[61] Yash Goyal, Ziyan Wu, Jan Ernst, Dhruv Batra, Devi Parikh, and Stefan Lee. Counterfactual visual explanations. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2376–2384, 2019.

[62] Hankyeol Lee, Gawon Seo, Wonseok Choi, Geunyoung Jung, Kyungwoo Song, and Jiyoung Jung. Enhancing visual classification using comparative descriptors. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5274–5283, 2025. doi: 10.1109/WACV61041.2025.00515.

[63] Sachit Menon and Carl Vondrick. Visual classification via description from large language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[64] Thomas Berg and Peter N. Belhumeur. POOF: Part-based one-vs.-one features for fine-grained categorization, face verification, and attribute estimation. In *2013 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 955–962, 2013. doi: 10.1109/CVPR.2013.128.

[65] Abhimanyu Dubey, Otkrist Gupta, Pei Guo, Ramesh Raskar, Ryan Farrell, and Nikhil Naik. Pairwise confusion for fine-grained visual classification. In *Computer Vision – ECCV 2018*, Lecture Notes in Computer Science, pages 71–88. Springer, 2018. doi: 10.1007/978-3-030-01258-8_5.

[66] Abhimanyu Dubey, Otkrist Gupta, Ramesh Raskar, and Nikhil Naik. Maximum-entropy fine-grained classification. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[67] Chang-Bin Zhang, Peng-Tao Jiang, Qibin Hou, Yunchao Wei, Qi Han, Zhen Li, and Ming-Ming Cheng. Delving deep into label smoothing. *IEEE Transactions on Image Processing*, 30:5984–5996, 2021. doi: 10.1109/TIP.2021.3089942.

[68] Jiaqi Wang, Wenwei Zhang, Yuhang Zang, Yuhang Cao, Jiangmiao Pang, Tao Gong, Kai Chen, Ziwei Liu, Chen Change Loy, and Dahua Lin. Seesaw loss for long-tailed instance segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9690–9699, 2021. doi: 10.1109/CVPR46437.2021.00957.

[69] Guolei Sun, Hisham Cholakkal, Salman Khan, Fahad Shahbaz Khan, and Ling Shao. Fine-grained recognition: Accounting for subtle differences between similar classes. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(07):12047–12054, 2020. doi: 10.1609/aaai.v34i07.6882.

[70] Lezi Wang, Ziyan Wu, Srikrishna Karanam, Kuan-Chuan Peng, Rajat Vikram Singh, Bo Liu, and Dimitris N. Metaxas. Sharpen focus: Learning with attention separability and consistency. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 512–521, 2019. doi: 10.1109/ICCV.2019.00060.

[71] Xue Zhang, Zehua Sheng, and Hui-Liang Shen. FocusNet: Classifying better by focusing on confusing classes. *Pattern Recognition*, 129:108709, 2022. doi: 10.1016/j.patcog.2022.108709.

[72] Peiqin Zhuang, Yali Wang, and Yu Qiao. Learning attentive pairwise interaction for fine-grained classification. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(07):13130–13137, 2020. doi: 10.1609/aaai.v34i07.7016.

[73] Han-Jia Ye, Hexiang Hu, De-Chuan Zhan, and Fei Sha. Few-shot learning via embedding adaptation with set-to-set functions. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8805–8814, 2020. doi: 10.1109/CVPR42600.2020.00883.

[74] Hongyang Li, David Eigen, Samuel Dodge, Matthew Zeiler, and Xiaogang Wang. Finding task-relevant features for few-shot learning by category traversal. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–10, 2019. doi: 10.1109/CVPR.2019.00009.

[75] Ruibing Hou, Hong Chang, Bingpeng Ma, Shiguang Shan, and Xilin Chen. Cross attention network for few-shot classification. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, 2019.

[76] Carl Doersch, Ankush Gupta, and Andrew Zisserman. CrossTransformers: spatially-aware few-shot transfer. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020.

[77] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025.

[78] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.

[79] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.

[80] Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science, pages 709–727. Springer, 2022. doi: 10.1007/978-3-031-19827-4_41.

[81] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016. doi: 10.1109/CVPR.2016.90.

[82] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11966–11976, 2022. doi: 10.1109/CVPR52688.2022.01167.

[83] Mingxing Tan and Quoc V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 6105–6114, 2019. URL https://proceedings.mlr.press/v97/tan19a.html.

[84] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image

recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[85] Francesca Murabito, Concetto Spampinato, Simone Palazzo, Daniela Giordano, Konstantin Pogorelov, and Michael Riegler. Top-down saliency detection driven by visual classification. *Computer Vision and Image Understanding*, 172:67–76, 2018. doi: 10.1016/j.cviu.2018.03.005.

[86] Yabin Zhang, Hui Tang, and Kui Jia. Fine-grained visual categorization using meta-learning optimization with sample selection of auxiliary data. In *Computer Vision – ECCV 2018*, Lecture Notes in Computer Science, pages 241–256, 2018. doi: 10.1007/978-3-030-01237-3_15.

[87] Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Joan Puigcerver, Jessica Yung, Sylvain Gelly, and Neil Houlsby. Big transfer (BiT): General visual representation learning. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, pages 491–507, 2020. doi: 10.1007/978-3-030-58558-7_29.

[88] Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and dogs. In *2012 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3498–3505, 2012. doi: 10.1109/CVPR.2012.6248092.

[89] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 5389–5400, 2019.

[90] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

[91] Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using Bayesian binning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 29(1), 2015. doi: 10.1609/aaai.v29i1.9602.

[92] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[93] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.

[94] Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[95] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: A good closed-set classifier is all you need. In *International Conference on Learning Representations (ICLR)*, 2022.

[96] Chaofan Chen, Oscar Li, Chaofan Tao, Alina Jade Barnett, Jonathan Su, and Cynthia Rudin. This looks like that: Deep learning for interpretable image recognition. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[97] Meike Nauta, Jörg Schlötterer, Maurice van Keulen, and Christin Seifert. PIP-Net: Patch-based intuitive prototypes for interpretable image classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2744–2753, 2023. doi: 10.1109/CVPR52729.2023.00269.

[98] Mohammad Reza Taesiri, Giang Nguyen, and Anh Nguyen. Visual correspondence-based explanations improve AI robustness and human-AI team accuracy. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pages 34287–34301, 2022. doi: 10.52202/068431-2485.

[99] Baifeng Shi, Trevor Darrell, and Xin Wang. Top-down visual attention from analysis by synthesis. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2102–2112, 2023. doi: 10.1109/CVPR52729.2023.00209.

[100] Yujie Zhu. CLIP-guided label-free discriminative region scoring for fine-grained classification, 2026.

[101] Siyi Liu, Hanjun Yang, Chenchen Zhang, Xiaorong Zhu, Xinyu Zuo, Lisheng Duan, Haijin Liang, Jin Ma, Junfu Pu, and Yongqi Zhang. From saliency to discriminability: Rank-preserving visual token pruning for VLM rerankers, 2026.

[102] Tianzhu Ye, Li Dong, Yuqing Xia, Yutao Sun, Yi Zhu, Gao Huang, and Furu Wei. Differential transformer. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://openreview.net/forum?id=OvoCm1gGhN.

[103] Lanyun Zhu, Tianrun Chen, Jianxiong Yin, Simon See, and Jun Liu. Learning gabor texture features for fine-grained recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1621–1631, 2023. doi: 10.1109/ICCV51070.2023.00156.

[104] Abhra Chaudhuri, Massimiliano Mancini, Zeynep Akata, and Anjan Dutta. Relational proxies: Emergent relationships as fine-grained discriminators. In *Advances in Neural Information Processing Systems*, volume 35, pages 31145–31157, 2022. doi: 10.52202/068431-2258.

[105] Ardhendu Behera, Zachary Wharton, Pradeep R. P. G. Hewage, and Asish Bera. Context-aware attentional pooling (CAP) for fine-grained visual classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 929–937, 2021. doi: 10.1609/aaai.v35i2.16176.

[106] Asish Bera, Zachary Wharton, Yonghuai Liu, Nik Bessis, and Ardhendu Behera. SR-GNN: Spatial relation-aware graph neural network for fine-grained image categorization. *IEEE Transactions on Image Processing*, 31:6017–6031, 2022. doi: 10.1109/TIP.2022.3205215.

[107] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.